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## Performance of Bayesian linear regression in a model with mismatch

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In this paper we analyse, for a model of linear regression with Gaussian covariates, the high-dimensional limit of the performance of a Bayesian estimator given by the mean of a log-concave posterior distribution with Gaussian prior. Although high-dimensional analysis of Bayesian estimators has been previously studied for Bayesian-optimal linear regression where the correct posterior is used for inference, much less is known when there is a mismatch. Here we consider a model in which the responses are known to be generated as linear combinations of the covariates but the distribution of the ground-truth regression coefficients and the Gaussian noise's variance are unknown. This regression task can be rephrased as a statistical mechanics model known as the Gardner spin glass, an analogy that we exploit. Using a leave-one-out approach we characterize the mean square error for the regression coefficients. We also derive the log-normalizing constant of the posterior. Similar models have been studied by Shcherbina and Tirozzi and by Talagrand, but our arguments are much more straightforward. An interesting consequence of our analysis is that in the quadratic loss case, the performance of the Bayesian estimator is independent of a global 'temperature' hyperparameter and matches the ridge estimator: sampling and optimizing are equally good. Instead, for the absolute value loss, there is an optimal finite temperature to select, which allows the Bayesian estimator to beat the corresponding M-estimator.

**Keywords:** high-dimensional Bayesian inference; statistical mechanics; disordered systems.

### 1. Introduction and set-up

Linear regression with random covariates in the high-dimensional regime, where both the number of data points and their dimension scale proportionally, is a paradigmatic problem for modern inference. It has been studied mostly from two complementary perspectives: (i) in Bayesian statistics where, due to important simplifications that occur in this setting, analyses have focused on the case of the estimator obtained as the expectation of the 'true' posterior distribution of the regression coefficients [5–7, 32]. This is usually referred to as the *Bayesian-optimal estimator*. (ii) Another large body of literature has studied instead the performance of optimization procedures in the context of robust statistics and M-estimation,

where an estimator is obtained as the mode of a log-concave posterior measure or, equivalently, as the minimizer of a convex cost function [1–3, 10, 11, 14, 17, 18, 21, 22, 36–38, 40, 41].

Our work focuses on Bayesian inference outside the optimal setting, for which much less is known. The lack of perfect knowledge of the model generating the data forces one to make certain inaccurate prior assumptions on the distribution of the regression coefficients as well as the form of the model. This is, in a sense, more realistic and of practical interest. One natural setting for this is to consider Bayesian linear regression with some sources of mismatch. Due to the lack of certain simplifying symmetries inherent to the optimal setting of inference, known as the ‘Nishimori identities’ in physics [9, 15, 29], even a small mismatch can result in several theoretical challenges that are highly amenable to mathematical and statistical studies.

### 1.1 Our model

In the present work we aim to study a mismatched linear regression model that is as follows. Let  $x^* = (x_i^*)_{i \leq N}$  be an unknown ground-truth of regression coefficients. Assume that

$$\lim_{N \rightarrow \infty} \mathbb{E} \left| \frac{\|x^*\|^2}{N} - \gamma \right|^2 = 0$$

for some deterministic  $\gamma \geq 0$ . Let  $\alpha > 0$  be a fixed parameter and let  $M = \lfloor \alpha N \rfloor^1$ . Consider a matrix  $G = (g_{ki})_{k \leq M, i \leq N}$  with i.i.d. standard normal entries. The columns in  $G$  are understood as random covariate vectors, while the rows correspond to independent samples. The responses  $y = (y_k)_{k \leq M}$  are obtained according to the linear model

$$y_k = S_k(x^*) + z_k \tag{1.1}$$

for  $S_k(x^*) := (\bar{G}x^*)_k$ , where  $\bar{G} := N^{-1/2}G$  and the variables  $z = (z_k)_{k \leq M} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \Delta_*)$  for some  $\Delta_* > 0$ , which gives the strength of the noise. Here we will consider the regression task of recovering  $x^*$  from the data  $\mathcal{D} := (y, \bar{G})$  with two sources of mismatch:

1. The distribution of  $x^*$  is assumed to be unknown. Thus, we take the prior to be that of a Gaussian vector with independent coordinates of mean 0 and variance  $1/(2\kappa) > 0$ .
2. In the distribution generating the data (1.1), the noise is given by i.i.d. Gaussian random variables of variance  $\Delta_*$ . But its distribution is also assumed to be unknown. We therefore consider a factorized log-concave noise distribution with density proportional to  $\prod_{k \leq M} \exp u(S_k(x) - y_k)$ , where  $u$  is a non-positive concave function on  $\mathbb{R}$  satisfying some further technical conditions given in Assumption 1.

Under these sources of mismatch, the Bayesian posterior distribution is given by

$$P(x \in A \mid \mathcal{D}) = \frac{1}{Z(\mathcal{D})} \int_A \exp \left( \sum_{k \leq M} u(S_k(x) - y_k) - \kappa \|x\|^2 \right) dx \tag{1.2}$$

<sup>1</sup> To be more precise, we might take  $M = \alpha N + o(N)$ . But tracking this correction makes no difference.

for any measurable subset  $A$  of  $\mathbb{R}^N$ , where  $Z(\mathcal{D})$  is the normalization constant, also called the partition function in the terminology of statistical physics. Our aim is to study the performance of the Bayesian estimator  $\hat{x}$  defined through the mean of this mismatched Bayesian posterior distribution:

$$\hat{x}(\mathcal{D}) := \int_{\mathbb{R}^N} x P(dx | \mathcal{D}). \quad (1.3)$$

## 1.2 Main results and contributions

Our main contributions are establishing the asymptotics of the log-normalizing constant and the performance of estimator (1.3). Informally, we prove that, whenever the four-dimensional system (3.2)–(3.5) presented below admits a unique solution, the log-normalizing constant converges to some non-random quantity  $F(q, \rho)$ , that is,

$$\lim_{N \rightarrow \infty} \mathbb{E} \left| \frac{1}{N} \ln Z(\mathcal{D}) - F(q, \rho) \right|^2 = 0;$$

and the mean square error converges to some non-random quantity  $q$ , that is,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E} \|\hat{x} - x^*\|_2^2 = q. \quad (1.4)$$

Here  $(q, \rho)$  is the unique critical point of the real-valued function  $F$  with domain over the set  $\{(q, \rho) : 0 \leq q \leq \rho < \infty\}$  defined in (3.6).

There are two major findings that follow from our results. First of all, somewhat surprisingly (1.4) implies that the performance of the Bayesian estimator with Gaussian prior is insensitive to the statistical properties of the coefficients  $x^*$  other than its empirical second moment. The second major finding concerns our main running example, which plays a special role due to its analytical tractability and significance in statistics. This example corresponds to the model where the function  $u(s)$  takes the form  $-s^2/(2\Delta)$  for some  $\Delta > 0$ . Notice that this model is still mismatched since  $\Delta$  is not necessarily equal to the  $\Delta_*$  appearing in the model generating the data. When we convert the posterior distribution into a Gibbs measure parametrized by the *inverse temperature*  $\beta$  we discover that, for quadratic  $u$ , the mean square error is independent of the inverse temperature  $\beta$  in the high-dimensional limit. This means that when the performance is evaluated using a square loss, Bayesian regression with Gaussian prior and ridge regression have the same performance. In other words, sampling and optimizing are equivalent in this setting. One may argue that the closedness between the mean and the mode of the posterior distribution with Gaussian prior is mainly due to the isotropy of the covariates combined with the quadratic nature of the loss. This intuition makes perfect sense in the classical regime, where a lot of data are accessible ( $M \gg N$ ) when maximum likelihood estimation is optimal [44]. But in the high-dimensional regime where both  $M$  and  $N$  are large and comparable, this feature becomes much less evident. To the best of our knowledge, this fact was not rigorously obtained prior to our work. Exploring the generality of this equivalence between optimization and sampling procedures is an interesting open direction for future research. This equivalence is, however, not true for all loss functions even if the covariates are isotropic (Gaussian). As a counter example, we study a second example that is a soft version of the absolute value loss. By numerically analysing our resulting fixed point equations for the mean square error in this case, we show that there exists an optimal finite  $\beta$  value that yields a minimum value of the error,

strictly smaller than when  $\beta \rightarrow \infty$ . Therefore, for this loss (and most probably a large class of losses), the Bayes estimator with fine-tuned temperature is indeed better than the corresponding M-estimator as one may expect. See [2, 3] for related connections between M-estimation and Bayesian inference from statistical mechanics heuristics.

The approach in this paper relies primarily on a connection between our regression task and a generalized version of the Shcherbina–Tirozzi (ST) mean-field spin glass model [35]. We study the limits of the corresponding log-partition function and some critical quantities called the overlaps. By utilizing the so-called smart path method, analogous results were intensively explored by Talagrand [39] assuming that  $u$  grows at most linearly. Although the machinery therein relies on quite heavy analysis, the advantage of our approach is methodological. In addition to being able to relax the growth condition for  $u$  from linear to quadratic, our analysis is also rather simple and self-contained. It is based on methods coming from the mathematical physics of mean-field spin glasses: mainly a rigorous version of the cavity method [30, 39] also called the leave-one-out method in statistics (see Appendices C and D for details). We expect for the simplicity of our approach to allow us to extend the analysis to generalized linear models in mismatched settings, for which similar results exist only in the Bayesian-optimal setting [6]. This is left for future work.

### 1.3 Related works

In recent years, many works on mismatched regression have appeared: on non-rigorous statistical mechanics techniques such as the replica method [24, 25] like, e.g. in [2]; on the analysis of approximate message-passing algorithms, like in [11, 17, 22, 36] and on Gordon’s convex min-max theorem [40, 41]. All these approaches strongly rely on the fact that the estimator is obtained as a unique minimizer of a convex function and therefore do not generalize easily to mismatched Bayesian settings. The leave-one-out method was also used in [18, 19] but, like in the aforementioned references, only for the analysis of (non-Bayesian) M-estimators.

Complementary to our set-up is a recent work [26] on the mismatched Bayesian linear regression that derives conditions under which the ‘naive mean-field approximation’ to the log-partition function is valid, and provides an infinite-dimensional variational formula for it (see [31] for a related setting). Finally, Bayesian linear regression has also been considered recently in some different high-dimensional scaling regimes than the one presented here: in settings with very sparse ground-truth vectors of coefficients with a vanishing fraction of non-zero entries. Reference [13] studies in detail the properties of the posterior distribution with sparsity inducing priors in this case. In this very sparse regime, with the number of sampled responses much smaller than the covariates’ dimension, intriguing ‘all-or-nothing’ phase transitions occur where the recovery error jumps from almost zero to its maximum value at a certain threshold [8, 20, 23, 27, 28, 33, 34]. For a recent review on the Bayesian approach to high-dimensional inference, see [4].

### 1.4 Organization

The paper is organized as follows. In Section 2, we explain the link between the high-dimensional regression problem presented here and the ST model. In Section 3, we first state two results on the free energy and overlaps in the ST model and then state our main results on the regression task with a particular emphasis on the classical example of the Bayesian regression with quadratic loss and prior, but also a soft version of absolute value loss. Appendices B–D are devoted to establishing results concerning the ST model. More explicitly, Appendix B establishes a number of concentration properties for the overlaps and free energy in the ST model. Appendix C shows how to obtain certain fixed-point equations

for the ‘order parameters’ of the model based on the concentration results in Appendix B by utilizing the cavity method and simple Gaussian integration by parts. Appendix D proves a closed-form expression for the limiting log-partition function of the posterior in a straightforward manner due to the convergence results for the order parameters. Finally, our results on the regression task are established in Appendix 3.

## 2. Notations used

In view of our regression task, the posterior measure (1.2) can be viewed as a Gibbs measure of the spin glass model with Hamiltonian:

$$\tilde{H}_N(x) := \sum_{k \leq M} u \left( \frac{1}{\sqrt{N}} \sum_{i \leq N} g_{ki}(x_i - x_i^*) - z_k \right) - \kappa \|x\|^2,$$

that is,

$$P(dx \mid \mathcal{D}) = \tilde{G}_N(dx) := \frac{e^{\tilde{H}_N(x)} dx}{\int e^{\tilde{H}_N(x')} dx'}. \quad (2.1)$$

One of the key quantities in our analysis will be the free energy of the model defined according to

$$\tilde{F}_N := \frac{1}{N} \log \int_{\mathbb{R}^N} \exp \tilde{H}_N(x) dx.$$

In the terminology of the large deviation theory [16, 42], it is the scaled cumulant log-moment generating function associated with the energy per variable  $\tilde{H}_N(x)/N$ . For a given realization of the randomness  $(G, z)$  defining an instance of the model, we will refer to each element of a sequence  $(x^l)_{l \geq 1}$  of independently drawn samples from the Gibbs measure as *replicas*. Denote by  $\langle \cdot \rangle^\sim$  the Gibbs expectation associated with the Hamiltonian  $\tilde{H}_N$ . In particular, from (1.3), we readily see that  $\hat{x} = \langle x \rangle^\sim$ .

For  $h \in \mathbb{R}$ , we will consider an auxiliary model with Hamiltonian  $H_N = (H_N(\sigma))_{\sigma \in \mathbb{R}^N}$  defined by

$$H_N(\sigma) := \sum_{k \leq M} u(S_k(\sigma) + z_k) - h \sum_{i \leq N} \sigma_i - \kappa \|\sigma\|^2 \quad (2.2)$$

and its corresponding free energy by

$$F_N := \frac{1}{N} \ln \int_{\mathbb{R}^N} \exp H_N(\sigma) d\sigma. \quad (2.3)$$

These are nothing more than the Hamiltonian and free energy of the generalized ST model [35]. When  $z_k \equiv 0$  and  $u$  grows at most linearly, the Hamiltonian  $H_N$  is known as the original ST model, extensively studied in [39]. However, in the setting of the present paper, we allow  $u$  to grow quadratically (see Assumption 1 for details).

As before, for a given realization of  $(G, z)$ , each element of  $(\sigma^l)_{l \geq 1}$  is an independently drawn sample from

$$G_N(d\sigma) := \frac{\exp H_N(\sigma) d\sigma}{\int \exp H_N(\sigma') d\sigma'}$$

and will be referred as a *replica*.

Denote by  $\langle \cdot \rangle$  the Gibbs expectation of  $(\sigma^\ell)_{\ell \geq 1}$ . Two key quantities for our analysis will be the *overlaps* between replicas defined according to

$$R_{1,1} := \frac{\|\sigma^1\|^2}{N} \quad \text{and} \quad R_{1,2} := \frac{\sigma^1 \cdot \sigma^2}{N}. \quad (2.4)$$

Additionally, we also define a pair of ‘conjugate overlaps’ in terms of the following two auxiliary quantities:

$$A(\sigma^\ell) := \left( u' \left( S_k(\sigma^\ell) + z_k \right) \right)_{k \leq M} \quad \text{and} \quad B(\sigma^\ell) := \left( u'' \left( S_k(\sigma^\ell) + z_k \right) \right)_{k \leq M}. \quad (2.5)$$

For convenience, we will sometimes write  $A^\ell = (A_k^\ell)_{k \leq M}$ ,  $B^\ell = (B_k^\ell)_{k \leq M}$ ,  $S_k^\ell$  instead of  $A(\sigma^\ell)$ ,  $B(\sigma^\ell)$  or  $S_k(\sigma^\ell)$ . Whenever the replica index  $\ell$  is omitted, it is assumed to be equal to 1. The conjugate overlaps are then defined by

$$Q_{1,1} := \frac{1}{N} \sum_{k \leq M} \left( (A_k)^2 + B_k \right) \quad \text{and} \quad Q_{1,2} := \frac{1}{N} \langle A^1, A^2 \rangle. \quad (2.6)$$

As one shall see, the vector  $(R_{1,1}, R_{1,2}, Q_{1,1}, Q_{1,2})$  of order parameters is the ‘correct’ one, in the sense that all the statistical analysis can be asymptotically expressed solely in terms of the limits of these quantities and the model at the ‘macroscopic level’ is completely determined by them.

In the rest of the paper we will let  $\gamma_N := \|x^*\|^2/N$  and  $h = h_N := 2\kappa\sqrt{\gamma_N}$ . From the derivations in Appendix A we have

$$\mathbb{E} \tilde{F}_N = -\kappa \mathbb{E} \gamma_N + \mathbb{E} F_N \quad \text{and} \quad \mathbb{E} \|\hat{x} - x^*\|^2 = \mathbb{E} \|\langle x \rangle^\sim - x^*\|^2 = \mathbb{E} \|\langle \sigma \rangle\|^2, \quad (2.7)$$

where the Hamiltonian  $H_N$  in  $F_N$  is evaluated in the random field  $h = h_N$ .

### 3. Main results

Owing to equations (2.7), to understand our regression problem, we may study the limiting free energy and overlaps corresponding to the generalized ST model first. Thus, the first part of our main results consists of an expression for the limiting free energy and the convergence of the overlaps in the generalized ST model defined in (2.2). Throughout the rest of this paper, the following assumption on the function  $u$  is in force.



ASSUMPTION 1 (Growth condition). The function  $u(s)$  is concave, non-positive and there exists a constant  $d > 0$  such that

$$\max \left( |u(0)|, |u'(0)|, \|u''\|_\infty, \|u'''\|_\infty, \|u^{(4)}\|_\infty \right) \leq d$$

and

$$|u'(s)| \leq d(1 + \sqrt{|u(s)|}).$$

REMARK 1. The concavity of  $u$  is conceptually an important assumption, which ensures that our posterior distribution is a log-concave measure and, as a result, the overlaps are concentrated under the Gibbs measure  $G_N$ . While the other assumptions are purely technical, overall they weaken the settings considered in [39]. In particular, in this reference  $u$  is allowed to grow at most linearly and thus its results do not apply to our main example. This is because the strategy used in [39] for proving overlap concentration is based on a concentration-of-measure-like result and requires a uniform control on the first and second derivatives of  $u$ . In contrast, our proof uses a combination of free energy concentration and Brascamp–Lieb’s inequality. This alternative strategy permits to remove the requirement of the uniform bound on the first derivative of  $u$  allowing for quadratic growth.

To formulate our limiting free energy, we introduce

$$\begin{aligned} \bar{F}(q, \rho, r, \bar{r}) = & \alpha \mathbb{E} \ln \mathbb{E}_\xi \exp u \left( \tilde{z} \sqrt{\Delta_* + q} + \xi \sqrt{\rho - q} \right) \\ & + \frac{1}{2} \left( \frac{r + h^2}{2\kappa + r - \bar{r}} - \ln(2\kappa + r - \bar{r}) + r q - \bar{r} \rho + \ln 2\pi \right) \end{aligned} \quad (3.1)$$

for  $0 \leq q < \rho$  and  $0 \leq \bar{r} \leq r$ , where  $\tilde{z}, \xi$  are independent  $\mathcal{N}(0, 1)$ . If we let  $\theta = \tilde{z} \sqrt{\Delta_* + q} + \xi \sqrt{\rho - q}$ , using Gaussian integration by parts with respect to  $\tilde{z}$  and  $\xi$  implies that the critical points of the function  $\bar{F}$  must satisfy the following system of equations:

$$q = \Psi(r, \bar{r}) := \frac{r + h^2}{(2\kappa + r - \bar{r})^2}, \quad (3.2)$$

$$\rho = \bar{\Psi}(r, \bar{r}) := \frac{1}{2\kappa + r - \bar{r}} + \frac{r + h^2}{(2\kappa + r - \bar{r})^2}, \quad (3.3)$$

$$r = \Phi(q, \rho) := \alpha \mathbb{E} \left( \frac{\mathbb{E}_\xi u'(\theta) \exp u(\theta)}{\mathbb{E}_\xi \exp u(\theta)} \right)^2, \quad (3.4)$$

$$\bar{r} = \bar{\Phi}(q, \rho) := \alpha \mathbb{E} \frac{\mathbb{E}_\xi (u''(\theta) + u'(\theta)^2) \exp u(\theta)}{\mathbb{E}_\xi \exp u(\theta)}. \quad (3.5)$$

REMARK 2. As will be clear in the proofs, these parameters represent the limiting values as  $N \rightarrow \infty$  of certain physical quantities of the system. The tightness of these quantities along with the fact, which we will establish later on, that the limits of convergent subsequences satisfy equations (3.2)–(3.5) readily proves the existence of solutions. However, the question of uniqueness is much more difficult in general,

as it involves proving regularity properties of this four-dimensional system of nonlinear equations. In the important case of quadratic  $u$  this problem is overcome by finding a solution to these equations by explicitly integrating the right-hand sides of (3.4)–(3.5).

In addition, for any  $0 \leq q < \rho$ , we consider the function

$$F(q, \rho) = \alpha \mathbb{E} \ln \mathbb{E}_\xi \exp u \left( \tilde{z} \sqrt{\Delta_* + q} + \xi \sqrt{\rho - q} \right) + \frac{1}{2} \left( \ln(\rho - q) + h^2(\rho - q) + \frac{\rho}{\rho - q} - 2\kappa\rho + \ln 2\pi \right). \quad (3.6)$$

Observe that if  $(q, \rho, r, \bar{r})$  satisfies (3.2)–(3.5), then  $\tilde{F}(q, \rho, r, \bar{r})$  and  $F(q, \rho)$  match each other as can be seen by rewriting (3.2) and (3.3) as

$$2\kappa + r - \bar{r} = \frac{1}{\rho - q} \quad \text{and} \quad r + h^2 = \frac{q}{(\rho - q)^2}$$

and plugging these into the second line in (3.1) after writing  $rq - \bar{r}\rho = -r(\rho - q) + (r - \bar{r})\rho$ . Here, if we express  $r, \bar{r}$  in terms of  $q, \rho$  in (3.4)–(3.5), the pair  $(q, \rho)$  is also a critical point of  $F$ . Our first main result states that the limiting free energy in the ST model is equal to  $F(q, \rho)$  as long as (3.2)–(3.5) have a unique solution.

**THEOREM 1.** For any  $\kappa > 0$ ,  $h \in \mathbb{R}$  and  $\Delta_* \geq 0$ , if (3.2)–(3.5) have a unique solution  $(q, \rho, r, \bar{r})$ , then

$$\lim_{N \rightarrow \infty} \mathbb{E} |F_N - F(q, \rho)|^2 = 0.$$

The next result establishes the convergence of the overlaps (2.4) and (2.6).

**THEOREM 2.** For any  $\kappa > 0$ ,  $h \in \mathbb{R}$  and  $\Delta_* \geq 0$ , if (3.2)–(3.5) have a unique solution  $(q, \rho, r, \bar{r})$ , then under  $\mathbb{E}(\cdot)$ ,

$$R_{1,1} \xrightarrow{L^2} \rho, \quad R_{1,2} \xrightarrow{L^2} q, \quad Q_{1,1} \xrightarrow{L^2} \bar{r} \quad \text{and} \quad Q_{1,2} \xrightarrow{L^2} r.$$

From these and with the help of the identities (2.7), we can now compute the limiting free energy and the mean square error associated with our regression task.

**THEOREM 3.** Assume that

$$\lim_{N \rightarrow \infty} \mathbb{E} |\gamma_N - \gamma|^2 = 0 \quad (3.7)$$

for some deterministic  $\gamma \geq 0$ . For  $\kappa > 0$ ,  $h = 2\kappa\sqrt{\gamma}$  and  $\Delta_* \geq 0$ , if (3.2)–(3.5) have a unique solution  $(q, \rho, r, \bar{r})$ , then

$$\lim_{N \rightarrow \infty} \mathbb{E} |\tilde{F}_N - (-\kappa\gamma + F(q, \rho))|^2 = 0 \quad (3.8)$$



and

$$\lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E} \|\hat{x} - x^*\|^2 = \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E} \|\langle x \rangle^\sim - x^*\|^2 = q. \quad (3.9)$$

EXAMPLE 1 (Quadratic loss). Let  $u(s) = -s^2/(2\Delta)$ , where  $\Delta > 0$  is not necessarily equal to the true parameter  $\Delta_*$ . Obviously  $u$  satisfies Assumption 1. More importantly, it can be checked that (3.2)–(3.5) have a unique solution given by

$$\begin{cases} q = \frac{c^2(h^2(\Delta+c)^2 + \Delta_*\alpha)}{(\Delta+c)^2 - \alpha c^2}, \\ \rho = q + c, \\ r = \frac{\alpha(\Delta_* + q)}{(\Delta + \rho - q)^2}, \\ \bar{r} = r - \frac{\alpha}{\Delta + \rho - q} \end{cases} \quad (3.10)$$

for

$$c := \frac{\sqrt{(2\kappa\Delta + \alpha - 1)^2 + 8\kappa\Delta} - (2\kappa\Delta + \alpha - 1)}{4\kappa}.$$

To see this, note that (3.4) and (3.5) can be written as

$$r = \frac{\alpha(\Delta_* + q)}{(\Delta + \rho - q)^2} \quad \text{and} \quad \bar{r} = r - \frac{\alpha}{\Delta + \rho - q}. \quad (3.11)$$

Take  $a = \rho - q$  and  $a' = r - \bar{r}$ . From the difference between (3.2) and (3.3) and that of (3.4) and (3.5), we arrive at

$$\begin{cases} aa' + 2\kappa a = 1, \\ aa' + \Delta a' = \alpha, \end{cases} \quad (3.12)$$

from which we can eliminate  $a'$  to get a single equation for  $a$ ,

$$2\kappa a^2 + (2\kappa\Delta + \alpha - 1)a - \Delta = 0.$$

It is easy to see that the only non-negative solution to this equation is  $a = c$  and thus,  $a' = \alpha/(\Delta + c)$ . Finally, using these values of  $a$  and  $a'$  and the second equation of (3.11) leads to

$$q = \frac{c^2(h^2(\Delta + c)^2 + \Delta_*\alpha)}{(\Delta + c)^2 - \alpha c^2}. \quad (3.13)$$

Note that  $q$  is positive since (3.12) implies that  $aa' < 1$  and  $aa' < \alpha$ , which ensures that  $(\Delta + c)^2/c^2 = (aa')^2 < \alpha$ . From this  $q$  and  $(\rho, r, \bar{r})$  can then be uniquely determined through the second to the fourth equations in (3.10).

Surprisingly, if we parametrize the posterior distribution by an inverse temperature  $\beta > 0$ , the original Gibbs measure  $\tilde{G}_N$  in (2.1) becomes

$$\tilde{G}_{N,\beta}(dx) = \frac{e^{\beta \tilde{H}_N(x)} dx}{\int e^{\beta \tilde{H}_N(x')} dx'},$$

which means that if  $\Delta$  and  $\kappa$  are replaced by  $\Delta/\beta$  and  $\beta\kappa$  in the original model, then the resulting constant  $q_\beta$  remains the same as the original  $q$  for any inverse temperature  $\beta > 0$ . At the same time, formula (3.13) matches the one for ridge regression at zero temperature (see, e.g. equation (26) in [40]). This non-trivial fact is due to the rotational invariance of the model following from the isotropy of the covariates. But as mentioned already in the introduction, this is not a priori evident in the high-dimensional regime we consider here, where  $M$  and  $N$  are comparable. It would be interesting to explore in future research for which settings the posterior mean and mode (i.e. sampling and optimizing) yield the same reconstruction performance.

EXAMPLE 2 (Soft absolute value).

Define, for every  $\epsilon > 0$ , the function  $u_\epsilon : \mathbb{R} \mapsto \mathbb{R}$  according to

$$u_\epsilon(s) = -\sqrt{s^2 + \epsilon}.$$

This function is a soft approximation of  $-|\cdot|$ . Notice that

$$u'_\epsilon(s) = -\frac{s}{\sqrt{s^2 + \epsilon}} \quad \text{and} \quad u''_\epsilon(s) = -\frac{\epsilon}{(s^2 + \epsilon)^{3/2}}.$$

It is then straightforward to check that for every  $\epsilon > 0$  there is a  $d > 0$  such that Assumption 1 holds. As in the previous example, we can use the results of the paper to find the corresponding fixed point equations for this potential when we also introduce an inverse temperature parameter  $\beta > 0$ . In contrast to the quadratic potential, the fixed point equations for this case do not have a closed form but can be nonetheless solved numerically. In Fig. 1 we present the dependency of the asymptotic mean square error, as given by Theorem 3, as a function of the inverse temperature  $\beta > 0$  with two different noise variances. Although the uniqueness requirement of the theorem cannot be definitely checked numerically, in every group of simulations with a given set of parameters only one solution of the fixed point equations was observed.

Two conclusions can be drawn from these graphs. The first one being that, when the loss is non-quadratic, there might be a non-trivial behaviour of the Bayesian estimator performance as a function of the inverse temperature. This directly implies that the analyses covering the performance at zero temperature do not cover the results of the present work. A second important remark is that there appears to be an optimal finite temperature at which the resulting estimator achieves a minimal mean square error and this temperature seems to be strictly positive in the cases studied. This means that with a proper choice of temperature, the Bayesian estimator improves the performance of the zero temperature M-estimator, and this despite the presence of mismatch in the posterior.

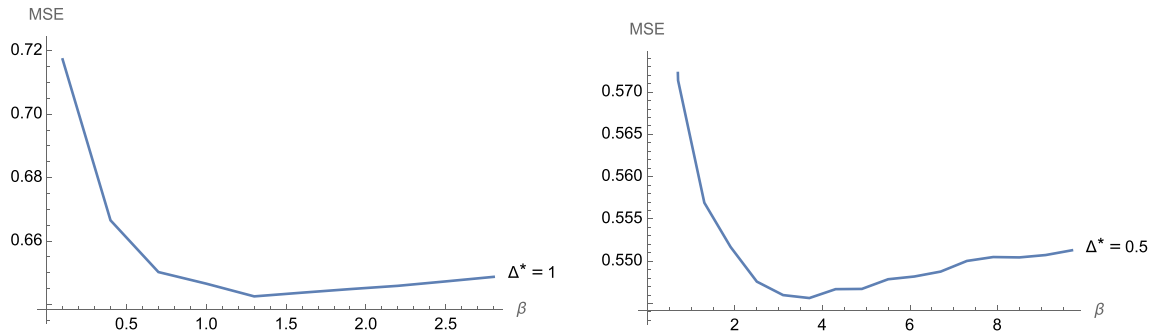


FIG. 1. Mean square error as a function of the inverse temperature for the soft absolute value loss  $u_\epsilon$  for two values of the noise variance: on the left  $\Delta_* = 1$ , and on the right  $\Delta_* = 0.5$ . The other parameters of the model take the values ( $\kappa = 0.5, \gamma = 1, \alpha = 1$ ) in both cases.

#### 4. Outline of the proofs

The general strategy of the proofs relies on first establishing the concentration of the free energy and the relevant order parameters (i.e. ‘replica symmetry’ in physics parlance) through concentration inequalities for log-concave measures; this is done in Appendix B. The proofs there involve having a tight control over the norm of samples from the Gibbs measure and introducing an appropriate perturbation to the Hamiltonian to prove the desired concentrations.

Using the replica symmetry of the Gibbs measure, in Appendix C the two pairs of fixed point equations for the asymptotic values of the means of the order parameters are derived. One pair of equations is derived by means of a cavity/leave-an-observation-out argument and the second pair is obtained using Gaussian integration by parts. This gives the proof of Theorem 2.

Finally, Theorem 1 is proved in Appendix D using the convergence of the order parameters towards the unique solution to the fixed point equations together with an interpolation argument. Finally, the proof of Theorem 3, which requires extending the previous theorems to the case where the deterministic parameter  $h$  is replaced by a random one  $h_N$ , involves another interpolation argument and some concentration inequalities; it is given in Appendix E.

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### A. Connection with the ST model

The connection between the ST and our models relies on a number of change of variables. First of all, by letting  $w := x - x^*$ , we may write

$$\begin{aligned} \int_{\mathbb{R}^N} \exp \tilde{H}_N(x) dx &= \int_{\mathbb{R}^N} \exp \left( \sum_{k \leq M} u \left( \frac{1}{\sqrt{N}} \sum_{i \leq N} g_{ki} w_i - z_k \right) - \kappa \|w + x^*\|^2 \right) dw \\ &= \exp(-\kappa \|x^*\|^2) \int_{\mathbb{R}^N} \exp \left( \sum_{k \leq M} u \left( \frac{1}{\sqrt{N}} \sum_{i \leq N} g_{ki} w_i - z_k \right) \right. \\ &\quad \left. - 2\kappa \sum_{i \leq N} x_i^* w_i - \kappa \|w\|^2 \right) dw. \end{aligned}$$

To simplify this integral, take an orthogonal matrix  $O \in \mathcal{O}(N)$  such that  $Ox^* = N^{-1/2} \|x^*\| \mathbf{1}_N$ . Note that conditionally on  $x^*$ ,

$$\begin{aligned} \mathbb{E}(GO^\top)_{k\ell}(GO^\top)_{k'\ell'} &= \mathbb{E} \left( \sum_{i \leq N} g_{ki} O_{\ell i} \right) \left( \sum_{i \leq N} g_{k'i} O_{\ell' i} \right) \\ &= \delta_{kk'} \sum_{i \leq N} O_{\ell i} O_{\ell' i} = \delta_{kk'} (OO^\top)_{\ell\ell'} = \delta_{kk'} \delta_{\ell\ell'}, \end{aligned}$$

which implies that  $GO^\top \sim G$  conditionally on  $x^*$ . Note also that  $\langle x^*, O^\top a \rangle = N^{-1/2} \|x^*\| \langle \mathbf{1}_N, a \rangle$  and  $\|O^\top a\| = \|a\|$ . From these and using the change of variable  $\sigma := Ow$ ,

$$\begin{aligned} \int_{\mathbb{R}^N} \exp \left( \sum_{k \leq M} u \left( \frac{1}{\sqrt{N}} \sum_{i \leq N} g_{ki} w_i - z_k \right) - 2\kappa \sum_{i \leq N} x_i^* w_i - \kappa \|w\|^2 \right) dw \\ = \int_{\mathbb{R}^N} \exp \left( \sum_{k \leq M} u \left( (\tilde{G}O^\top \sigma)_k - z_k \right) - 2\kappa \langle x^*, O^\top \sigma \rangle - \kappa \|O^\top \sigma\|^2 \right) d\sigma \\ = \int_{\mathbb{R}^N} \exp \left( \sum_{k \leq M} u \left( (\tilde{G}O^\top \sigma)_k - z_k \right) - \frac{2\kappa \|x^*\|}{\sqrt{N}} \sum_{i \leq N} \sigma_i - \kappa \|\sigma\|^2 \right) d\sigma, \end{aligned} \quad (\text{A.1})$$

which implies, from the rotational invariance of  $G$  mentioned above and the symmetry of  $z_k$ , that

$$\begin{aligned} \mathbb{E} \ln \int_{\mathbb{R}^N} \exp \left( \sum_{k \leq M} u \left( \frac{1}{\sqrt{N}} \sum_{i \leq N} g_{ki} w_i - z_k \right) - 2\kappa \sum_{i \leq N} x_i^* w_i - \kappa \|w\|^2 \right) dw \\ = \mathbb{E} \ln \int_{\mathbb{R}^N} \exp \left( \sum_{k \leq M} u \left( (\tilde{G}\sigma)_k + z_k \right) - \frac{2\kappa \|x^*\|}{\sqrt{N}} \sum_{i \leq N} \sigma_i - \kappa \|\sigma\|^2 \right) d\sigma. \end{aligned}$$

From this we then conclude

$$\mathbb{E}\tilde{F}_N = -\kappa\mathbb{E}\gamma_N + \frac{1}{N}\mathbb{E}\ln \int_{\mathbb{R}^N} \exp \left( \sum_{k \leq M} u(S_k(\sigma) + z_k) - 2\kappa\sqrt{\gamma_N} \sum_{i \leq N} \sigma_i - \kappa\|\sigma\|^2 \right) d\sigma \quad (\text{A.2})$$

for  $\gamma_N := \|x^*\|^2/N$ .

If now we randomize  $h = h_N = 2\kappa\sqrt{\gamma_N}$  in the free energy  $F_N$  of the ST model independently of all other randomness, then we readily get

$$\mathbb{E}\tilde{F}_N = -\kappa\mathbb{E}\gamma_N + \mathbb{E}F_N$$

and similarly,

$$\mathbb{E}\|\hat{x} - x^*\|^2 = \mathbb{E}\|\langle x \rangle^\sim - x^*\|^2 = \mathbb{E}\|\langle \sigma \rangle\|^2. \quad (\text{A.3})$$

## B. Concentration

In this section we will prove a number of useful concentration results for overlaps and the free energy. For this, it will be convenient to restate the growth conditions in Assumption 1 in a more general way, more suitable for the proof. Take  $n \geq 1$  and consider two functions  $\phi, \psi : \mathbb{R}^n \mapsto \mathbb{R}$ , which together with the function  $u : \mathbb{R} \mapsto \mathbb{R}$  satisfy the following set of conditions. Namely, we suppose that there exist  $d > 0$  and  $\eta_0 > 0$  such that the following holds:

(G-1) The function  $u(s)$  satisfies

$$0 \geq u(s) \geq -d(1 + s^2) \quad \text{and} \quad |u'(s)| \leq d(1 + |s|). \quad (\text{B.1})$$

(G-2) The functions  $\phi, \psi : \mathbb{R}^n \mapsto \mathbb{R}$  satisfy

$$\max(|\phi(s) - \phi(s')|, |\psi(s) - \psi(s')|) \leq d(1 + \|s\| + \|s'\|)\|s - s'\|, \quad (\text{B.2})$$

$$\max(|\phi(s)|, |\psi(s)|) \leq d(1 + \|s\|^2), \quad (\text{B.3})$$

$$\max_{1 \leq \ell, \ell' \leq n} (|\partial_{s_\ell s_{\ell'}} \phi(s)|, |\partial_{s_\ell s_{\ell'}} \psi(s)|) \leq d. \quad (\text{B.4})$$

(G-3) The following inequality holds:

$$\sup_{y \in \mathbb{R}^n} \left( \sum_{\ell \leq n} u(y^\ell) + \eta_0 |\psi(y^1, \dots, y^n)| \right) \leq \alpha d.$$

The last assumption (G-3) will be used in an equivalent form:

$$\max_{y^1, \dots, y^n \in \mathbb{R}^M} \sum_{k \leq M} \left( \sum_{\ell \leq n} u(y_k^\ell) + \eta_0 |\psi(y_k^1, \dots, y_k^n)| \right) \leq Nd. \quad (\text{B.5})$$

Also, the assumptions (B.2) and (B.3) may look redundant given (B.4), but we state them for convenience.

We will later specialize these assumptions for the following two specific choices:

- For  $n = 1$ :  $\phi(s) = s^2$  and  $\psi(s) = u''(s) + u'(s)^2$ ,
- For  $n = 2$ :  $\phi(s_1, s_2) = s_1 s_2$  and  $\psi(s_1, s_2) = u'(s_1)u'(s_2)$ .



One can easily see that Assumption 1 on  $u$  implies (G-1)–(G-3) in both cases, by possibly increasing the value of  $d$ .

For  $\sigma^1, \dots, \sigma^n \in \mathbb{R}^N$  denote  $\vec{\sigma} = (\sigma^1, \dots, \sigma^n) \in \mathbb{R}^{Nn}$  and set  $\|\vec{\sigma}\| = (\|\sigma^1\|^2 + \dots + \|\sigma^n\|^2)^{1/2}$ . Consider any  $\lambda$  and  $\eta$  satisfying  $|\lambda|, |\eta| \leq \min(\eta_0, \kappa/(2d))$ . For any  $\sigma \in \mathbb{R}^N$  recall  $S_k(\sigma) := (\bar{G}\sigma)_k$  and define

$$X_N(\sigma) := \sum_{k \leq M} u(S_k(\sigma) + z_k).$$

For  $\vec{\sigma} \in \mathbb{R}^{Nn}$ , we define an Hamiltonian between ‘coupled replicas’  $(\sigma^\ell)_{\ell \leq n}$ :

$$X_{N,\lambda,\eta}(\vec{\sigma}) := \sum_{\ell \leq n} X_N(\sigma^\ell) + h \sum_{\ell \leq n} \sum_{i \leq N} \sigma_i^\ell - \kappa \|\vec{\sigma}\|^2 + \lambda N R(\vec{\sigma}) + \eta N Q(\vec{\sigma}), \quad (\text{B.6})$$

where the generalized overlaps are

$$R(\vec{\sigma}) := \frac{1}{N} \sum_{i \leq N} \phi(\sigma_i^1, \dots, \sigma_i^n), \quad (\text{B.7})$$

$$Q(\vec{\sigma}) := \frac{1}{N} \sum_{k \leq M} \psi(S_k(\sigma^1) + z_k, \dots, S_k(\sigma^n) + z_k), \quad (\text{B.8})$$

for functions  $\phi$  and  $\psi$  verifying assumptions (G-1)–(G-3). Define the (non-averaged) free energy associated with  $X_{N,\lambda,\eta}$  by

$$F_N(\lambda, \eta) := \frac{1}{N} \ln Z_{N,\lambda,\eta} \quad \text{where} \quad Z_{N,\lambda,\eta} := \int_{\mathbb{R}^{Nn}} \exp X_{N,\lambda,\eta}(\vec{\sigma}) d\vec{\sigma}.$$

Note that assumptions (B.1), (B.2) and (B.5) as well as  $|\lambda|, |\eta| \leq \min(\eta_0, \kappa/(2d))$  ensure that this quantity is a.s. finite in  $G$  and  $z = (z_k)$ . Denote by  $\langle \cdot \rangle_{\lambda,\eta}$  the expectation with respect to the Gibbs measure proportional to  $\exp X_{N,\lambda,\eta}(\vec{\sigma})$ :

$$\langle g(\vec{\sigma}) \rangle_{\lambda,\eta} := \frac{1}{Z_{N,\lambda,\eta}} \int_{\mathbb{R}^{Nn}} g(\vec{\sigma}) \exp X_{N,\lambda,\eta}(\vec{\sigma}) d\vec{\sigma}.$$

From now on, when we say that ‘there exist constants  $\varepsilon, K, K', D, L$ , etc.’, these constants are functions of all the parameters of the model  $(d, n, \alpha, \Delta_*, \kappa, h, \eta_0)$ , which are bounded uniformly over compacts. As before,  $\mathbb{E}$  is the expectation with respect to the Gaussian variables  $(G, z)$ .

**PROPOSITION B1.** Under the above assumptions, there exists some  $K > 0$  such that

$$\mathbb{E} \left\langle \left| R(\vec{\sigma}) - \mathbb{E} \langle R(\vec{\sigma}) \rangle_{0,0} \right|^2 \right\rangle_{0,0} \leq \frac{K}{\sqrt{N}} \quad \text{and} \quad \mathbb{E} \left\langle \left| Q(\vec{\sigma}) - \mathbb{E} \langle Q(\vec{\sigma}) \rangle_{0,0} \right|^2 \right\rangle_{0,0} \leq \frac{K}{\sqrt{N}}. \quad (\text{B.9})$$

Moreover, for any  $k \geq 1$ , there exists some  $C_k > 0$  such that

$$\mathbb{E} \langle |R(\vec{\sigma})|^k \rangle_{0,0} \leq C_k \quad \text{and} \quad \mathbb{E} \langle |Q(\vec{\sigma})|^k \rangle_{0,0} \leq C_k.$$

In the rest of this section, we will establish Proposition B.1. To fix our notation, for any real  $M \times N$ -matrix  $A$ , define the Frobenius and operator norms, respectively, as

$$\|A\| := \left( \sum_{k \leq M} \sum_{i \leq N} A_{ki}^2 \right)^{1/2} \quad \text{and} \quad \|A\|_{op} := \sup_{y \in \mathbb{R}^N: \|y\| \leq 1} \|Ay\|.$$

Recall  $\|\cdot\|$  applied to a vector is the  $L_2$  norm and note that  $\|A\|_{op} \leq \|A\|$ .

### B.1 Control of the self-overlap

Let  $\bar{z} := z/\sqrt{N}$  and recall also that  $\bar{G} := G/\sqrt{N}$ .

LEMMA B1. There exist  $\varepsilon > 0$  and  $K \geq 1$  such that if  $t, |\lambda|, |\eta| \leq \varepsilon$  then

$$\langle \exp t \|\bar{\sigma}\|^2 \rangle_{\lambda, \eta} \leq \exp KN(\|\bar{z}\|^2 + \|\bar{G}\|_{op}^2 + 1).$$

*Proof.* From (B.3),

$$|R(\bar{\sigma})| \leq \frac{d}{N} \sum_{i \leq N} \left( 1 + \sum_{\ell \leq n} (\sigma_i^\ell)^2 \right) = \frac{d}{N} (N + \|\bar{\sigma}\|^2) \quad (\text{B.10})$$

and

$$|Q(\bar{\sigma})| \leq \frac{d}{N} \left( M + \sum_{\ell \leq n} \sum_{k \leq M} (S_k(\sigma^\ell) + z_k)^2 \right) \leq \frac{d}{N} \left( M + 2\|\bar{G}\|_{op}^2 \|\bar{\sigma}\|^2 + 2n\|z\|^2 \right). \quad (\text{B.11})$$

By (B.10), if  $t + d|\lambda| \leq \kappa/2$ , then

$$t\|\bar{\sigma}\|^2 - \kappa\|\bar{\sigma}\|^2 + \lambda NR(\bar{\sigma}) \leq |\lambda|dN - \frac{\kappa}{2}\|\bar{\sigma}\|^2,$$

which, together with (B.5) for  $|\eta| \leq \eta_0$ , implies that

$$\int \exp \left( t\|\bar{\sigma}\|^2 + X_{N, \lambda, \eta}(\bar{\sigma}) \right) d\bar{\sigma} \leq \exp N \left( d + d|\lambda| + \frac{nh^2}{2\kappa} + \frac{n}{2} \ln \frac{2\pi}{\kappa} \right), \quad (\text{B.12})$$

where we used that for any  $a > 0$  and  $b \in \mathbb{R}$ ,

$$\int \exp(-ax^2 + bx) dx = \left( \frac{\pi}{a} \right)^{1/2} \exp \frac{b^2}{4a}. \quad (\text{B.13})$$

Next, since by (B.1)

$$u(S_k(\sigma) + z_k) \geq -d(1 + 2S_k(\sigma)^2 + 2z_k^2),$$

we have

$$\sum_{k \leq M} u(S_k(\sigma) + z_k) \geq -d \left( M + 2\|\bar{G}\|_{op}^2 \|\sigma\|^2 + 2\|z\|^2 \right).$$

Combining this with (B.10) and (B.11), we obtain that

$$X_{N, \lambda, \eta}(\bar{\sigma}) \geq -\tilde{T}\|\bar{\sigma}\|^2 - N\tilde{T}' + h \sum_{\ell \leq n} \sum_{i \leq N} \sigma_i^\ell,$$

where

$$\tilde{T} := \kappa + 2d(1 + |\eta|)\|\bar{G}\|_{op}^2 + d|\lambda| > 0 \quad \text{and} \quad \tilde{T}' := d(\alpha n + |\lambda| + \alpha|\eta|) + 2nd(1 + |\eta|)\|\bar{z}\|^2 > 0.$$

Consequently, from (B.13),

$$Z_{N,\lambda,\eta} \geq \exp nN \left( -\frac{\tilde{T}'}{n} + \frac{h^2}{4\tilde{T}} + \frac{1}{2} \ln \frac{\pi}{\tilde{T}} \right) \geq \exp nN \left( -\frac{\tilde{T}'}{n} - \frac{1}{2} \ln \tilde{T} \right),$$

which together with the fact that  $\ln x \leq x + 1$  for all  $x > 0$  implies that

$$\frac{1}{Z_{N,\lambda,\eta}} \leq \exp nN \left( \frac{\tilde{T}'}{n} + \frac{1}{2}(\tilde{T} + 1) \right) \leq \exp K'N \left( \|\bar{z}\|^2 + \|\bar{G}\|_{op}^2 + 1 \right).$$

Our assertion follows by putting this and (B.12) together.  $\square$

We let  $I(A)$  be the indicator function for event  $A$ .

LEMMA B2. There exist constants  $\varepsilon > 0$  and  $K, K' > 0$  such that, for any  $|\lambda|, |\eta| \leq \varepsilon$ ,

$$\left\langle I \left( \|\bar{\sigma}\| \geq K' \sqrt{N(\|\bar{z}\|^2 + \|\bar{G}\|_{op}^2 + 1)} \right) \right\rangle_{\lambda,\eta} \leq \exp(-KN).$$

*Proof.* Recall the constants  $\varepsilon$  and  $K$  from the statement of Lemma B1. If  $t \leq \varepsilon$ , then for any  $a > 0$ ,

$$\langle I(\|\bar{\sigma}\| \geq a) \rangle_{\lambda,\eta} \leq \langle \exp t(\|\bar{\sigma}\|^2 - a^2) \rangle_{\lambda,\eta} \leq \exp(-a^2 t + KN(\|\bar{z}\|^2 + \|\bar{G}\|_{op}^2 + 1)).$$

Our proof is completed by taking  $a = \sqrt{2t^{-1}KN(\|\bar{z}\|^2 + \|\bar{G}\|_{op}^2 + 1)}$ .  $\square$

LEMMA B3. Let  $k \geq 1$ . There exist constants  $0 < \varepsilon < 1$  and  $K \geq 1$  such that, for any  $|\lambda|, |\eta| \leq \varepsilon$ ,

$$\langle \|\bar{\sigma}\|^{2k} \rangle_{\lambda,\eta} \leq KN^k \left( \|\bar{z}\|^{2k} + \|\bar{G}\|_{op}^{2k} + 1 \right). \quad (\text{B.14})$$

*Proof.* From Lemma B1,

$$\ln \langle \exp t\|\bar{\sigma}\|^2 \rangle_{\lambda,\eta} \leq KN(\|\bar{z}\|^2 + \|\bar{G}\|_{op}^2 + 1).$$

For any non-negative random variable  $X$ , by lemma 3.1.8 in [39],

$$\mathbb{E}X^k \leq 2^k \left( k^k + (\ln \mathbb{E} \exp X)^k \right).$$

Using this inequality with  $X = t\|\bar{\sigma}\|^2$ , we have

$$\begin{aligned} t^k \langle \|\bar{\sigma}\|^{2k} \rangle_{\lambda,\eta} &\leq 2^k \left( k^k + K^k N^k (\|\bar{z}\|^2 + \|\bar{G}\|_{op}^2 + 1)^k \right) \\ &\leq 2^k \left( k^k + 3^k K^k N^k (\|\bar{z}\|^{2k} + \|\bar{G}\|_{op}^{2k} + 1) \right) \\ &\leq C_k N^k \left( \|\bar{z}\|^{2k} + \|\bar{G}\|_{op}^{2k} + 1 \right). \end{aligned}$$

This finishes the proof.  $\square$

## B.2 Convexity and pseudo-Lipschitz property of the overlaps

LEMMA B4. If  $|\lambda| + \|\bar{G}\|_{op}^2 |\eta| \leq \kappa/(dn)$  then the following function is convex:

$$\vec{\sigma} \mapsto \frac{\kappa}{2} \|\vec{\sigma}\|^2 - \lambda NR(\vec{\sigma}) - \eta NQ(\vec{\sigma}). \quad (\text{B.15})$$

*Proof.* Recall the generalized overlap definitions (B.7) and (B.8). Note that for any  $\vec{y} = (y^1, \dots, y^n) \in \mathbb{R}^N \times \dots \times \mathbb{R}^N$ ,

$$\sum_{\ell, \ell' \leq n} \sum_{i, i' \leq N} \partial_{\sigma_i^\ell \sigma_{i'}^{\ell'}} \left( \sum_{j \leq N} \phi(\sigma_j^1, \dots, \sigma_j^n) \right) y_i^\ell y_{i'}^{\ell'} = \sum_{\ell, \ell' \leq n} \left( \sum_{i \leq N} \partial_{s_\ell s_{\ell'}} \phi(\sigma_i^1, \dots, \sigma_i^n) y_i^\ell y_i^{\ell'} \right).$$

Since, by (B.4),  $|\partial_{s_\ell s_{\ell'}} \phi(s)| \leq d$ , this is bounded in absolute value by

$$d \sum_{i \leq N} \left( \sum_{\ell \leq n} |y_i^\ell| \right)^2 \leq dn \|\vec{y}\|^2.$$

Similarly,

$$\begin{aligned} & \sum_{\ell, \ell' \leq n} \sum_{i, i' \leq N} \partial_{\sigma_i^\ell \sigma_{i'}^{\ell'}} \left( \sum_{k \leq M} \psi(S_k(\sigma^1) + z_k, \dots, S_k(\sigma^n) + z_k) \right) y_i^\ell y_{i'}^{\ell'} \\ &= \sum_{\ell, \ell' \leq n} \sum_{k \leq M} \partial_{s_\ell s_{\ell'}} \psi(S_k(\sigma^1) + z_k, \dots, S_k(\sigma^n) + z_k) \left( \sum_{i \leq N} \bar{g}_{ki} y_i^\ell \right) \left( \sum_{i' \leq N} \bar{g}_{ki'} y_{i'}^{\ell'} \right), \end{aligned}$$

and since, by (B.4),  $|\partial_{s_\ell s_{\ell'}} \psi(s)| \leq d$ , this is bounded in absolute value by

$$d \sum_{\ell, \ell'=1}^n |(\bar{G}y^\ell, \bar{G}y^{\ell'})| \leq d \|\bar{G}\|_{op}^2 \left( \sum_{\ell \leq n} \|y^\ell\| \right)^2 \leq dn \|\bar{G}\|_{op}^2 \|\vec{y}\|^2.$$

These complete our proof.  $\square$

LEMMA B5. We have that

$$|R(\vec{\sigma}) - R(\vec{\sigma}')| \leq \frac{d}{N} \left( \sqrt{N} + \|\vec{\sigma}\| + \|\vec{\sigma}'\| \right) \|\vec{\sigma} - \vec{\sigma}'\|$$

and

$$|Q(\vec{\sigma}) - Q(\vec{\sigma}')| \leq \frac{d}{N} \left( \sqrt{M} + 2\sqrt{n}\|z\| + \|\bar{G}\|_{op} \|\vec{\sigma}\| + \|\bar{G}\|_{op} \|\vec{\sigma}'\| \right) \|\bar{G}\|_{op} \|\vec{\sigma} - \vec{\sigma}'\|.$$

*Proof.* For simplicity, for  $\vec{\sigma} = (\sigma^1, \dots, \sigma^n)$ , we denote  $\sigma(i) = (\sigma_i^1, \dots, \sigma_i^n)$ . From (B.2),

$$\begin{aligned}
 |R(\vec{\sigma}) - R(\vec{\sigma}')| &\leq \frac{d}{N} \sum_{i \leq N} (1 + \|\sigma(i)\| + \|\sigma'(i)\|) \|\sigma(i) - \sigma'(i)\| \\
 &\leq \frac{d}{N} \left( \sum_{i \leq N} (1 + \|\sigma(i)\| + \|\sigma'(i)\|)^2 \right)^{1/2} \left( \sum_{i \leq N} \|\sigma(i) - \sigma'(i)\|^2 \right)^{1/2} \\
 &\leq \frac{d}{N} \left( \left( \sum_{i \leq N} 1 \right)^{1/2} + \left( \sum_{i \leq N} \|\sigma(i)\|^2 \right)^{1/2} + \left( \sum_{i \leq N} \|\sigma'(i)\|^2 \right)^{1/2} \right) \|\vec{\sigma} - \vec{\sigma}'\| \\
 &\leq \frac{d}{N} (\sqrt{N} + \|\vec{\sigma}\| + \|\vec{\sigma}'\|) \|\vec{\sigma} - \vec{\sigma}'\|,
 \end{aligned}$$

where in the third line we used the Minkowski inequality. Again, by (B.2),

$$\begin{aligned}
 |Q(\vec{\sigma}) - Q(\vec{\sigma}')| &\leq \frac{d}{N} \sum_{k \leq M} \left( 1 + \left( \sum_{\ell \leq n} |S_k(\sigma^\ell) + z_k|^2 \right)^{1/2} + \left( \sum_{\ell \leq n} |S_k(\sigma'^\ell) + z_k|^2 \right)^{1/2} \right) \\
 &\quad \times \left( \sum_{\ell \leq n} S_k(\sigma^\ell - \sigma'^\ell)^2 \right)^{1/2} \\
 &\leq \frac{d}{N} \left( \sum_{k \leq M} \left( 1 + \left( \sum_{\ell \leq n} |S_k(\sigma^\ell) + z_k|^2 \right)^{1/2} + \left( \sum_{\ell \leq n} |S_k(\sigma'^\ell) + z_k|^2 \right)^{1/2} \right)^2 \right)^{1/2} \\
 &\quad \times \left( \sum_{k \leq M} \sum_{\ell \leq n} S_k(\sigma^\ell - \sigma'^\ell)^2 \right)^{1/2}.
 \end{aligned}$$

Here, since

$$\left( \sum_{k \leq M} \sum_{\ell \leq n} S_k(\sigma^\ell - \sigma'^\ell)^2 \right)^{1/2} = \left( \sum_{\ell \leq n} \|\tilde{G}(\sigma^\ell - \sigma'^\ell)\|^2 \right)^{1/2} \leq \|\tilde{G}\|_{op} \|\vec{\sigma} - \vec{\sigma}'\|,$$

and, similarly,

$$\left( \sum_{k \leq M} \sum_{\ell \leq n} |S_k(\sigma^\ell) + z_k|^2 \right)^{1/2} \leq \left( \sum_{k \leq M} \sum_{\ell \leq n} S_k(\sigma^\ell)^2 \right)^{1/2} + \left( \sum_{k \leq M} \sum_{\ell \leq n} z_k^2 \right)^{1/2} \leq \|\tilde{G}\|_{op} \|\vec{\sigma}\| + \sqrt{n}\|z\|,$$

we can again use the Minkowski inequality to obtain

$$|Q(\vec{\sigma}) - Q(\vec{\sigma}')| \leq \frac{d}{N} \left( \sqrt{M} + 2\sqrt{n}\|z\| + \|\bar{G}\|_{op}\|\vec{\sigma}\| + \|\bar{G}\|_{op}\|\vec{\sigma}'\| \right) \|\bar{G}\|_{op}\|\vec{\sigma} - \vec{\sigma}'\|.$$

□

LEMMA B6. For any  $k \geq 1$  there exists some  $C_k > 0$  such that

$$\mathbb{E} \left\langle |R(\vec{\sigma})|^k \right\rangle_{0,0} \leq C_k \quad \text{and} \quad \mathbb{E} \left\langle |Q(\vec{\sigma})|^k \right\rangle_{0,0} \leq C_k.$$

*Proof.* Let us denote by  $\vec{s} := \vec{\sigma} / \sqrt{N}$ . Note that from Lemma B5,

$$\begin{aligned} |R(\vec{\sigma})| &\leq |R(0)| + \frac{d}{N} \left( \sqrt{N} + \|\vec{\sigma}\| \right) \|\vec{\sigma}\| \leq d + d(1 + \|\vec{s}\|) \|\vec{s}\|, \\ |Q(\vec{\sigma})| &\leq |Q(0)| + \frac{d}{N} \left( \sqrt{M} + 2\sqrt{n}\|z\| + \|\bar{G}\|_{op}\|\vec{\sigma}\| \right) \|\bar{G}\|_{op}\|\vec{\sigma}\| \\ &\leq d(\alpha + n\|\vec{z}\|^2) + d \left( \sqrt{\alpha} + 2\sqrt{n}\|\vec{z}\| + \|\bar{G}\|_{op}\|\vec{s}\| \right) \|\bar{G}\|_{op}\|\vec{s}\|, \end{aligned}$$

where we used the assumption (B.3). Using Lemma B.3 with  $\lambda = \eta = 0$ , we see that there exists some  $C$  such that

$$\langle \|\vec{s}\|^{2k} \rangle_{0,0} \leq C \left( \|\vec{z}\|^{2k} + \|\bar{G}\|_{op}^{2k} + 1 \right).$$

By corollary 5.35 in [43],

$$\mathbb{P}(\|\bar{G}\|_{op} \geq 1 + \sqrt{\alpha} + t) \leq 2 \exp(-Nt^2/2), \quad (\text{B.16})$$

which implies that  $\mathbb{E}\|\bar{G}\|_{op}^m \leq C_m < \infty$  for any  $m \geq 1$  and some constant  $C_m$  independent of  $N$ . Similarly, by Gaussian concentration for Gaussian vectors,  $\mathbb{E}\|\vec{z}\|^m \leq C_m < \infty$ . Together with the above inequalities, this finishes the proof. □

### B.3 Concentration of the free energy

For clarity, here we will now denote  $F_N(\lambda, \eta)$  by  $F(G, z)$ . Let us start with the following result.

LEMMA B7. There exist constants  $\varepsilon > 0$  and  $K > 0$  such that for all  $|\lambda|, |\eta| \leq \varepsilon$ ,

$$|F(G, z) - F(G', z')| \leq K \left( \|\vec{z}\|^3 + \|\bar{G}\|_{op}^3 + \|\vec{z}'\|^3 + \|\bar{G}'\|_{op}^3 + 1 \right) \|(\bar{G} - \bar{G}', \vec{z} - \vec{z}')\|. \quad (\text{B.17})$$

*Proof.* First of all,

$$\partial_{g_{ki}} F(G, z) = \frac{1}{N^{3/2}} \sum_{\ell \leq n} \left\langle \left( u' \left( S_k(\sigma^\ell) + z_k \right) + \eta \partial_{s_\ell} \psi \left( S_k(\sigma^1) + z_k, \dots, S_k(\sigma^n) + z_k \right) \right) \sigma_i^\ell \right\rangle_{\lambda, \eta}.$$

Using that  $|u'(s)| \leq d(1 + |s|)$  by (B.1),

$$\begin{aligned}
 \sum_{k \leq M} \sum_{i \leq N} \left\langle \sum_{\ell \leq n} u' \left( S_k(\sigma^\ell) + z_k \right) \sigma_i^\ell \right\rangle_{\lambda, \eta}^2 &\leq \left\langle \left( \sum_{k \leq M} \sum_{\ell \leq n} u' \left( S_k(\sigma^\ell) + z_k \right)^2 \right) \sum_{i \leq N} \sum_{\ell \leq n} (\sigma_i^\ell)^2 \right\rangle_{\lambda, \eta} \\
 &\leq 3d \left\langle \left( \sum_{k \leq M} \sum_{\ell \leq n} \left( 1 + S_k(\sigma^\ell)^2 + z_k^2 \right) \right) \|\vec{\sigma}\|^2 \right\rangle_{\lambda, \eta} \\
 &\leq 3d \left\langle \left( nM + \|\vec{G}\|_{op}^2 \|\vec{\sigma}\|^2 + n\|z\|^2 \right) \|\vec{\sigma}\|^2 \right\rangle_{\lambda, \eta} \\
 &\leq KN^2 \left( \|\vec{z}\|^6 + \|\vec{G}\|_{op}^6 + 1 \right),
 \end{aligned}$$

where the last inequality used (B.14). Note that from (B.4), there exists a constant  $d'$  such that

$$|\partial_{s_\ell} \psi(s)| \leq d' (1 + \|s\|).$$

Using this inequality, we can argue similarly that

$$\begin{aligned}
 \sum_{k \leq M} \sum_{i \leq N} \left\langle \sum_{\ell \leq n} \partial_{x_\ell} \psi \left( S_k(\sigma^1) + z_k, \dots, S_k(\sigma^n) + z_k \right) \sigma_i^\ell \right\rangle_{\lambda, \eta}^2 \\
 \leq d'^2 \left\langle \left( \sum_{k \leq M} \sum_{\ell \leq n} \left[ 1 + \left( \sum_{\ell'=1}^n |S_k(\sigma^{\ell'}) + z_k|^2 \right)^{1/2} \right]^2 \right) \sum_{i \leq N} \sum_{\ell \leq n} (\sigma_i^\ell)^2 \right\rangle_{\lambda, \eta} \\
 \leq 4d'^2 n \left\langle \left( \sum_{k \leq M} \left[ 1 + \sum_{\ell'=1}^n (S_k(\sigma^{\ell'})^2 + z_k^2) \right] \right) \|\vec{\sigma}\|^2 \right\rangle_{\lambda, \eta} \\
 \leq 4d'^2 n \left\langle \left( M + \|\vec{G}\|_{op}^2 \|\vec{\sigma}\|^2 + n\|z\|^2 \right) \|\vec{\sigma}\|^2 \right\rangle_{\lambda, \eta} \\
 \leq K'N^2 \left( \|\vec{z}\|^6 + \|\vec{G}\|_{op}^6 + 1 \right).
 \end{aligned}$$

Putting these together yields that

$$\sum_{k \leq M} \sum_{i \leq N} |\partial_{g_{ki}} F(G, z)|^2 \leq \frac{K''}{N} \left( \|\vec{z}\|^6 + \|\vec{G}\|_{op}^6 + 1 \right).$$

In similar manner, we readily compute that

$$\partial_{z_k} F(G, z) = \frac{1}{N} \sum_{\ell \leq n} \left\langle u' \left( S_k(\sigma^\ell) + z_k \right) + \eta \partial_{x_\ell} \psi \left( S_k(\sigma^1) + z_k, \dots, S_k(\sigma^n) + z_k \right) \right\rangle_{\lambda, \eta}$$

and

$$\sum_{k \leq M} |\partial_{z_k} F(G, z)|^2 \leq \frac{K'''}{N} \left( \|\vec{z}\|^4 + \|\vec{G}\|_{op}^4 + 1 \right).$$



All together, we get that

$$\|\nabla F(G, z)\| \leq \frac{K_4}{\sqrt{N}} \left( \|\bar{z}\|^3 + \|\bar{G}\|_{op}^3 + 1 \right),$$

which implies (B.17).  $\square$

LEMMA B8. There exist constants  $\varepsilon > 0$  and  $K > 0$  such that, for all  $|\lambda|, |\eta| \leq \varepsilon$ ,

$$\text{Var} F_N(\lambda, \eta) \leq \frac{K}{N}.$$

*Proof.* For simplicity of notation, denote

$$y := (G, z), \quad f(y) := F(G, z) \quad \text{and} \quad r(y) := \|\bar{G}\|_{op}^3 + \|\bar{z}\|^3.$$

If we define  $\bar{y} := y/\sqrt{N}$ , then the above Lemma proves that

$$|f(y) - f(y')| \leq K (r(y) + r(y') + 1) \|\bar{y} - \bar{y}'\|.$$

Let  $L > 0$  be a constant to be chosen later. Since, for  $r(y), r(y') \leq L$ ,

$$f(y) \leq f(y') + K(2L + 1) \|\bar{y} - \bar{y}'\|$$

and we have equality for  $y' = y$ , if we define

$$\Phi(y) := \inf \{f(y') + K(2L + 1) \|\bar{y} - \bar{y}'\| : r(y') \leq L\},$$

then  $\Phi(y) = f(y)$  for all  $y$  such that  $r(y) \leq L$ . From this definition, it can be seen from the reverse triangle inequality that

$$|\Phi(y) - \Phi(y')| \leq K(2L + 1) \|\bar{y} - \bar{y}'\| = \frac{K(2L + 1)}{\sqrt{N}} \|y - y'\| \quad (\text{B.18})$$

and

$$|\Phi(y) - f(y)| \leq (|\Phi(y)| + |f(y)|) I(r(y) > L). \quad (\text{B.19})$$

By (B.18) and the Gaussian–Poincaré concentration inequality,

$$\mathbb{E} |\Phi(y) - \mathbb{E} \Phi(y)|^2 \leq \mathbb{E} \|\nabla \Phi(y)\|^2 \leq \frac{K^2(2L + 1)^2}{N}. \quad (\text{B.20})$$

Next, using (B.19) and the Cauchy–Schwarz inequality, we control

$$\begin{aligned} |\mathbb{E} f(y) - \mathbb{E} \Phi(y)| &\leq \mathbb{E} |f(y) - \Phi(y)| \\ &\leq \mathbb{E} (|\Phi(y)| + |f(y)|) I(r(y) > L) \\ &\leq \sqrt{2 (\mathbb{E} f(y)^2 + \mathbb{E} \Phi(y)^2) \mathbb{P}(r(y) \geq L)}. \end{aligned}$$

Recalling that  $r(y) = \|\bar{G}\|_{op}^3 + \|\bar{z}\|^3$ , by (B.16) and Gaussian concentration for  $\|\bar{z}\|$ , for large enough  $L > 0$ , we have  $\mathbb{P}(r(y) \geq L) \leq 4 \exp(-N)$ . On the other hand, by Lemma B7 with  $(G, z) = y$  and

$(G', z') = 0$  we have

$$\begin{aligned} f(y)^2 &\leq 2f(0)^2 + 2K^2 \left( \|\bar{z}\|^3 + \|\bar{G}\|_{op}^3 + 1 \right)^2 \frac{\|y\|^2}{N} \\ &\leq 2f(0)^2 + 2K^2 N^4 \left( \frac{\|\bar{z}\|^3}{N^{3/2}} + \frac{\|\bar{G}\|^3}{N^{3/2}} + \frac{1}{N^{3/2}} \right)^2 \left( \frac{\|\bar{G}\|^2}{N} + \frac{\|\bar{z}\|^2}{N} \right), \end{aligned}$$

and since  $|f(0)|$  does not depend on  $N$  and all the moments of  $\|\bar{z}\|$  and  $\|\bar{G}\|/\sqrt{N}$  are uniformly bounded, we have that  $\mathbb{E}f(y)^2 \leq K'N^4$ . Similarly,

$$\Phi(y)^2 \leq 2f(0)^2 + 2K^2 N(2L+1)^2 \left( \frac{\|\bar{G}\|^2}{N} + \frac{\|\bar{z}\|^2}{N} \right),$$

from which we conclude that  $\mathbb{E}\Phi(y)^2 \leq K''N$ . Finally, using the bound

$$\mathbb{E}|f(y) - \mathbb{E}f(y)|^2 \leq 3 \left( \mathbb{E}|f(y) - \Phi(y)|^2 + \mathbb{E}|\Phi(y) - \mathbb{E}\Phi(y)|^2 + |\mathbb{E}\Phi(y) - \mathbb{E}f(y)|^2 \right)$$

finishes the proof.  $\square$

#### B.4 Proof of Proposition B1

We will only consider the case of the overlap  $R(\vec{\sigma})$ , since the proof for  $Q(\vec{\sigma})$  is almost identical. In this section we will denote by  $K$  a constant that depends only on the parameters of the model, but can vary from one occurrence to the next. Let us consider the event

$$E_N := \left\{ \|\bar{G}\|_{op} \leq K \right\} \cap \left\{ \|\bar{z}\| \leq K \right\}.$$

By (B.16), we can and will choose  $K \geq 1$  so that the probability of the event  $E_N$  is at least  $1 - 4e^{-N}$ . From Lemma B6,

$$\mathbb{E} \left[ \left| R(\vec{\sigma}) - \mathbb{E}\langle R(\vec{\sigma}) \rangle_{0,0} \right|^2 \middle| I(E_N^c) \right] \leq 4 \left( \mathbb{E}\langle R(\vec{\sigma})^4 \rangle_{0,0} \mathbb{P}(E_N^c) \right)^{1/2} \leq K \exp(-N/2). \quad (\text{B.21})$$

Let  $F'_N$  be the  $\lambda$ -derivative of  $F_N$ . In the remainder we will control the same expectation on the event  $E_N$ , and we will use the convexity of  $F_N(\lambda, \eta)$  in  $\lambda$ , which implies that

$$\begin{aligned} \left| \langle R(\vec{\sigma}) \rangle_{0,0} - \mathbb{E}\langle R(\vec{\sigma}) \rangle_{0,0} \right| &= \left| F'_N(0,0) - \mathbb{E}F'_N(0,0) \right| \\ &\leq F'_N(\lambda, 0) - F'_N(-\lambda, 0) + \frac{1}{\lambda} (|F_N(\lambda, 0) - \mathbb{E}F_N(\lambda, 0)| \\ &\quad + |F_N(0, 0) - \mathbb{E}F_N(0, 0)| + |F_N(-\lambda, 0) - \mathbb{E}F_N(-\lambda, 0)|). \end{aligned} \quad (\text{B.22})$$

Take  $\varepsilon > 0$  such that, for  $|\lambda|, |\eta| \leq \varepsilon$ , the inequalities in Lemmas B2, B3 and B8 hold. By Lemma B8, the expectation of the square of the second line above can be bounded by  $K/(N\lambda^2)$ .

In order to control the first term  $F'_N(\lambda, 0) - F'_N(-\lambda, 0)$  above, our goal will be to control

$$\partial_{\lambda\lambda} F_N(\lambda, \eta) = N \left\langle |R(\vec{\sigma}) - \langle R(\vec{\sigma}) \rangle_{\lambda, \eta}|^2 \right\rangle_{\lambda, \eta}.$$

For simplicity of notation we denote  $\vec{s} := \vec{\sigma}/\sqrt{N}$ . Since, for  $|\lambda|, |\eta| \leq \varepsilon$ , the inequalities in Lemmas B2 and B3 hold, on the event  $E_N$ , for some  $L \geq 1$ ,

$$\left\langle I(\|\vec{s}\| \geq L) \right\rangle_{\lambda, \eta} \leq \exp(-KN) \quad \text{and} \quad \left\langle \|\vec{s}\|^{10} \right\rangle_{\lambda, \eta} \leq K. \quad (\text{B.23})$$

Also, by Lemma B5,

$$|R(\vec{\sigma}) - R(\vec{\sigma}')| \leq D(1 + \|\vec{s}\| + \|\vec{s}'\|) \|\vec{s} - \vec{s}'\|, \quad (\text{B.24})$$

where we also denoted  $\vec{s}' := \vec{\sigma}'/\sqrt{N}$ .

We now proceed as in the proof of Lemma B8. Take  $L$  as in (B.23) and  $D$  as in (B.24), denote  $r(\vec{\sigma}) := \|\vec{s}\|$  and let

$$\rho(\vec{\sigma}) := \inf\{R(\vec{\sigma}') + D(2L + 1)\|\vec{s} - \vec{s}'\| : r(\vec{\sigma}') \leq L\}.$$

Then, from (B.24),  $\rho(\vec{\sigma}) = R(\vec{\sigma})$  for all  $r(\vec{\sigma}) \leq L$ ,

$$|\rho(\vec{\sigma}) - \rho(\vec{\sigma}')| \leq D(2L + 1) \|\vec{s} - \vec{s}'\|, \quad (\text{B.25})$$

$$|\rho(\vec{\sigma}) - R(\vec{\sigma})| \leq (|\rho(\vec{\sigma})| + |R(\vec{\sigma})|) I(r(\vec{\sigma}) > L). \quad (\text{B.26})$$

Using (B.26) and the Cauchy–Schwarz and Minkowski inequalities,

$$\begin{aligned} \left\langle |\rho(\vec{\sigma}) - R(\vec{\sigma})|^2 \right\rangle_{\lambda, \eta} &\leq \left\langle (|\rho(\vec{\sigma})| + |R(\vec{\sigma})|)^2 I(r(\vec{\sigma}) > L) \right\rangle_{\lambda, \eta} \\ &\leq 2 \left( \left\langle \rho(\vec{\sigma})^4 \right\rangle_{\lambda, \eta}^{1/2} + \left\langle R(\vec{\sigma})^4 \right\rangle_{\lambda, \eta}^{1/2} \right) \left\langle I(r(\vec{\sigma}) > L) \right\rangle_{\lambda, \eta}^{1/2} \\ &\leq 2 \left( \left\langle \rho(\vec{\sigma})^4 \right\rangle_{\lambda, \eta}^{1/2} + \left\langle R(\vec{\sigma})^4 \right\rangle_{\lambda, \eta}^{1/2} \right) \exp(-KN), \end{aligned}$$

where in the last line we used the first equation in (B.23). Since  $R(0) = \phi(0)$ , by (B.24)

$$|R(\vec{\sigma})| \leq |\phi(0)| + D(1 + \|\vec{s}\|) \|\vec{s}\|.$$

Since  $\rho(0) = R(0) = \phi(0)$ , by (B.25),

$$|\rho(\vec{\sigma})| \leq |\phi(0)| + D(2L + 1) \|\vec{s}\|.$$

Therefore, the second equation in (B.23) implies that

$$\left\langle \rho(\vec{\sigma})^4 \right\rangle_{\lambda, \eta}^{1/2} + \left\langle R(\vec{\sigma})^4 \right\rangle_{\lambda, \eta}^{1/2} \leq K$$

and we get that

$$\left\langle |\rho(\vec{\sigma}) - R(\vec{\sigma})|^2 \right\rangle_{\lambda, \eta} \leq K \exp(-KN). \quad (\text{B.27})$$

By Lemma B4, on the event  $E_N$ , if  $|\lambda| + K|\eta| \leq \kappa/(dn)$  then function (B.15) is convex, and thus the Hamiltonian  $X_{N, \lambda, \eta}(\vec{\sigma})$  in (B.6) is strictly concave on  $\mathbb{R}^{Nn}$  and its Hessian is bounded by  $-\frac{\kappa}{2}\text{Id}$ . In this case, the Brascamp–Lieb inequality (see, e.g. [12, theorem 5.1]) implies that

$$\left\langle |\rho(\vec{\sigma}) - \langle \rho(\vec{\sigma}) \rangle_{\lambda, \eta}|^2 \right\rangle_{\lambda, \eta} \leq \frac{2}{\kappa} \left\langle \|\nabla \rho(\vec{\sigma})\|^2 \right\rangle_{\lambda, \eta} \leq \frac{K}{N},$$

where the last inequality used (B.25). Together with (B.27), this implies that, on the event  $E_N$ ,

$$\partial_{\lambda\lambda} F_N(\lambda, \eta) = N \left\langle \left| R(\vec{\sigma}) - \langle R(\vec{\sigma}) \rangle_{\lambda, \eta} \right|^2 \right\rangle_{\lambda, \eta} \leq K,$$

assuming as above that  $|\lambda|, |\eta| \leq \varepsilon$  and  $|\lambda| + K|\eta| \leq \kappa/(dn)$ . In particular,

$$\left\langle \left| R(\vec{\sigma}) - \langle R(\vec{\sigma}) \rangle_{0,0} \right|^2 \right\rangle_{0,0} \leq \frac{K}{N} \quad (\text{B.28})$$

and  $|F'_N(\lambda, 0) - F'_N(-\lambda, 0)| \leq K\lambda$ . Plugging this into (B.22) and, as we mentioned above, using Lemma B8 for the second term,

$$\mathbb{E} \left[ \left| \langle R(\vec{\sigma}) \rangle_{0,0} - \mathbb{E} \langle R(\vec{\sigma}) \rangle_{0,0} \right|^2 I(E_N) \right] \leq K \left( \lambda^2 + \frac{1}{N\lambda^2} \right) \leq \frac{K}{\sqrt{N}},$$

by optimizing over  $\lambda$ . Using (B.28) on the event  $E_N$  and then combining with (B.21) gives

$$\mathbb{E} \left\langle \left| R(\vec{\sigma}) - \mathbb{E} \langle R(\vec{\sigma}) \rangle_{0,0} \right|^2 \right\rangle_{0,0} \leq \frac{K}{\sqrt{N}}.$$

This proves the first inequality in (B.9). The proof of the second inequality is identical.  $\square$

### C. Fixed point equations

This section establishes the proof of Theorem 2. The key ingredient here is to show that equations (3.2)–(3.5) are satisfied by the subsequential limits of the overlaps, based on a cavity/leave-one-out argument and Gaussian integration by parts.

#### C.1 *M-cavity*

For every measurable function  $f : \mathbb{R}^{Nn} \mapsto \mathbb{R}$ , define the Gibbs expectation associated with the ‘cavity in  $M$ ’ by

$$\langle f \rangle_c := \frac{1}{Z_c^n} \int_{\mathbb{R}^{Nn}} \left( \prod_{\ell \leq n} d\sigma^\ell \right) f \exp \sum_{\ell \leq n} H_{N,c}(\sigma^\ell),$$

where

$$H_{N,c}(\sigma^\ell) := \sum_{k=2}^M u \left( S_k(\sigma^\ell) + z_k \right) + h \sum_{i \leq N} \sigma_i^\ell - \kappa \|\sigma^\ell\|^2$$

and  $Z_c$  is its corresponding partition function. In other words, the Hamiltonian here singles out the component  $u(S_M(\sigma) + z_M)$  from  $H_N(\sigma)$ . It is easy to check that for every measurable  $f : \mathbb{R}^{Nn} \mapsto \mathbb{R}$ ,

$$\langle f(\sigma^1, \dots, \sigma^n) \rangle = \frac{\langle f(\sigma^1, \dots, \sigma^n) \exp \sum_{\ell \leq n} u(S_1(\sigma^\ell) + z_1) \rangle_c}{\langle \exp \sum_{\ell \leq n} u(S_1(\sigma^\ell) + z_1) \rangle_c}, \quad (\text{C.1})$$

where  $\langle \cdot \rangle$  is the expectation with respect to the Gibbs measure proportional to  $\exp \sum_{\ell \leq n} H_N(\sigma^\ell)$  of the replicated system. The following two lemmas will play an essential role in establishing the fixed point equations in the next subsection.

LEMMA C1. For every  $k \geq 1$ , there exists some  $C_k > 0$  such that, uniformly on  $N$ ,

$$\mathbb{E}\langle |u(S_1(\sigma) + z_1)|^k \rangle \leq C_k.$$

*Proof.* Here we will denote  $u(S_1(\sigma) + z_1)$  by  $u_1 \leq 0$ . Because  $|u_1|^k$  is an increasing function of  $|u_1|$  and  $\exp(-|u_1|)$  is decreasing, the Harris inequality implies  $\langle |u_1|^k \exp(-|u_1|) \rangle_c \leq \langle |u_1|^k \rangle_c \langle \exp(-|u_1|) \rangle_c$ . As a consequence we have that

$$\mathbb{E}\langle |u_1|^k \rangle = \mathbb{E} \frac{\langle |u_1|^k \exp(-|u_1|) \rangle_c}{\langle \exp(-|u_1|) \rangle_c} \leq \mathbb{E}\langle |u_1|^k \rangle_c.$$

Furthermore, because the standard Gaussian vector  $g_1 := (g_{11}, \dots, g_{1N})$  is independent of  $\sigma$  under the cavity measure, by the rotational invariance of  $g_1$  we have that  $\langle g_1, \sigma \rangle \sim g \|\sigma\|$ , where  $g$  is an independent standard Gaussian variable, so under the cavity measure  $S_1(\sigma) \sim g \|\sigma\| / \sqrt{N} = g \sqrt{R_{1,1}}$ . Thus, by Assumption 1

$$\mathbb{E}\langle |u_1|^k \rangle_c \leq d^k \mathbb{E}\langle (1 + 2g^2 R_{1,1} + 2z_1^2)^k \rangle_c.$$

Finally, by the independence of  $R_{1,1}$  from  $g$  and  $z_1$  and Proposition B1, the right-hand side of this last equation is easily seen to be smaller than some constant that does not depend on  $N$ .  $\square$

LEMMA C2. There exist constants  $K, K' > 0$  such that

$$\left| \mathbb{E}\langle R_{1,1} \rangle_c - \mathbb{E}\langle R_{1,1} \rangle \right| \leq \frac{K}{\sqrt{N}} \quad \text{and} \quad \left| \mathbb{E}\langle R_{1,2} \rangle_c - \mathbb{E}\langle R_{1,2} \rangle \right| \leq \frac{K'}{\sqrt{N}}.$$

*Proof.* As in the previous lemma, we will denote  $u(S_1(\sigma) + z_1)$  by  $u_1$ . Let

$$H_{N,t}(\sigma) := H_{N,c}(\sigma) + t u_1(\sigma)$$

for  $t \in [0, 1]$ , and denote by  $\langle \cdot \rangle_t$  the expectation with respect to the associated Gibbs measure. We then clearly have that

$$\mathbb{E}\langle R_{1,1} \rangle_c = \mathbb{E}\langle R_{1,1} \rangle_{t=0} \quad \text{and} \quad \mathbb{E}\langle R_{1,1} \rangle = \mathbb{E}\langle R_{1,1} \rangle_{t=1}.$$

Along this interpolation,

$$\left| \frac{d}{dt} \mathbb{E}\langle R_{1,1} \rangle_t \right| = \left| \mathbb{E}\langle (R_{1,1} - \langle R_{1,1} \rangle_t) u_1 \rangle_t \right| \leq \sqrt{\mathbb{E}\langle (R_{1,1} - \langle R_{1,1} \rangle_t)^2 \rangle_t} \times \sqrt{\mathbb{E}\langle u_1^2 \rangle_t} \leq \frac{K}{\sqrt{N}} \times K',$$

for some constants  $K, K' > 0$  that do not depend on  $t \in [0, 1]$ . Here, the bound for the first factor follows from the Brascamp–Lieb inequality, while the bound on the second term is a generalization of Lemma C1. The proof for  $R_{1,2}$  is similar.  $\square$

## C.2 Consistency equations for the overlaps

Throughout this subsection, we will repeatedly use the concentration results for the generalized overlaps  $Q$  and  $R$  from Appendix B that are applicable to the overlaps  $Q_{1,1}$  using  $\psi(s) = u''(s) + u'(s)^2$ ,  $Q_{1,2}$  using  $\psi(s_1, s_2) = u'(s_1)u'(s_2)$ ,  $R_{1,1}$  using  $\phi(s) = s^2$  and  $R_{1,2}$  using  $\phi(s_1, s_2) = s_1 s_2$ . Denote

$$\rho_N := \mathbb{E}\langle R_{1,1} \rangle, \quad q_N := \mathbb{E}\langle R_{1,2} \rangle, \quad \bar{r}_N := \mathbb{E}\langle Q_{1,1} \rangle \quad \text{and} \quad r_N := \mathbb{E}\langle Q_{1,2} \rangle.$$

By Proposition B1, there exists some compact set  $K$  such that for all  $N \geq 1$ ,  $(\rho_N, q_N, \bar{r}_N, r_N)$  belongs to  $K$ . Thus, there exists a subsequence  $(N_m)_{m \geq 1}$  of system sizes along which

$$(\rho_m, q_m, \bar{r}_m, r_m) := (\rho_{N_m}, q_{N_m}, \bar{r}_{N_m}, r_{N_m}) \xrightarrow{m \rightarrow +\infty} (\rho, q, \bar{r}, r)$$

for some  $(\rho, q, \bar{r}, r)$  in  $K$ . For the rest of this section, we work with this convergent subsequence and we aim to show that  $(\rho, q, \bar{r}, r)$  satisfies (3.2)–(3.5) in Propositions C1 and C2. To begin with, we establish the following lemma.

LEMMA C3. Let, for  $n \geq 1$ ,  $f : \mathbb{R}^n \mapsto \mathbb{R}$  be continuous and bounded and

$$\theta_\ell := \sqrt{\rho - q} \xi_\ell + \sqrt{q + \Delta_*} \tilde{z} \quad \text{for } \ell \leq n, \quad (\text{C.2})$$

with  $\tilde{z}, (\xi_\ell)_{\ell \leq n}$  i.i.d. standard Gaussian variables. Under Assumption 1, we have that

$$\mathbb{E} \langle f(S_1(\sigma^1) + z_1, \dots, S_1(\sigma^n) + z_1) \rangle_c \xrightarrow{m \rightarrow +\infty} \mathbb{E} f(\theta_1, \dots, \theta_n).$$

*Proof.* Along the proof  $\ell$  and  $\ell'$  always belong to  $\{1, \dots, n\}$ . Recall  $S_1^\ell := S_1(\sigma^\ell)$ . Because  $f$  is continuous and bounded and because for fixed values of  $\sigma^1, \dots, \sigma^\ell$  the quantity  $(S_1^1 + z_1, \dots, S_1^\ell + z_1)$  is a Gaussian vector with covariance matrix  $\Sigma_m$  with entries  $\Sigma_{m, \ell \ell'} = R_{\ell, \ell'} + \Delta_*$ , we have that there exists some continuous and bounded  $g : \mathbb{R}^{n \times n} \mapsto \mathbb{R}$  such that

$$\mathbb{E} \langle f(S_1^1 + z_1, \dots, S_1^n + z_1) \rangle_c = \mathbb{E} \langle g(\Sigma_m) \rangle_c. \quad (\text{C.3})$$

By Proposition B1 and Lemma C2,  $\Sigma_m \xrightarrow{L^2} \Sigma$  as  $m \rightarrow +\infty$  where

$$\Sigma_{\ell \ell'} = (\rho + \Delta_*) I(\ell = \ell') + (q + \Delta_*) I(\ell \neq \ell').$$

Then, by the continuous mapping theorem,  $\mathbb{E} \langle g(\Sigma_m) \rangle_c \rightarrow g(\Sigma)$ . Observe that  $\Sigma = DD^\top$ , where

$$D_{kk'} = \sqrt{q + \Delta_*} I(k' = 1) + \sqrt{\rho - q} I(k' = k + 1) \in \mathbb{R}^{n \times n+1}.$$

We can then define i.i.d. standard Gaussian variables  $v := (\tilde{z}, \xi_1, \dots, \xi_n)$  so that  $\theta_\ell := (Dv)_\ell$  is (C.2), and then by reversing relation (C.3) we get  $g(\Sigma) = \mathbb{E} f(\theta_1, \dots, \theta_n)$ .  $\square$

PROPOSITION C1. The vector  $(\rho, q, \bar{r}, r)$  satisfies (3.4) and (3.5).

*Proof.* Let  $(\theta_\ell)_{\ell \leq n}$  as in (C.2),  $\theta = \theta_1$ ,  $\xi = \xi_1$  and  $a_\ell := S_1(\sigma^\ell) + z_1$ . Let  $K \geq 1$ . For each  $k \leq K$  define (we let  $\prod_{j=2}^1 (\dots) = 1$ )

$$f_k(a_1, \dots, a_k) := (u'(a_1)^2 + u''(a_1)) \exp u(a_1) \prod_{j=2}^k (1 - \exp u(a_j)).$$

By equation (C.1), the overlap definition (2.6), the spin symmetry and the geometric series (recall  $u(s) \leq 0$ ) we have that

$$\begin{aligned}\mathbb{E}\langle Q_{1,1} \rangle &= \alpha \mathbb{E}\langle u'(a_1)^2 + u''(a_1) \rangle \\ &= \alpha \mathbb{E} \frac{\langle (u'(a_1)^2 + u''(a_1)) \exp u(a_1) \rangle_c}{\langle \exp u(a_1) \rangle_c} \\ &= \alpha \sum_{k \leq K} \mathbb{E}\langle f_k(a_1, \dots, a_k) \rangle_c + \alpha \mathbb{E} \left( \langle u'(a_1)^2 + u''(a_1) \rangle (1 - \langle \exp u(a_1) \rangle_c)^K \right).\end{aligned}\quad (\text{C.4})$$

Note that by Assumption 1 the first term is the mean of a continuous and bounded function of  $a_1, \dots, a_K$ . Then by Lemma C3 we have that

$$\sum_{k \leq K} \mathbb{E}\langle f_k(a_1, \dots, a_k) \rangle_c \xrightarrow{m \rightarrow +\infty} \sum_{k \leq K} \mathbb{E}f_k(\theta_1, \dots, \theta_k).$$

The second term can be bounded in the following way:

$$\begin{aligned}& \left| \mathbb{E} \left( \langle u'(a_1)^2 + u''(a_1) \rangle (1 - \langle \exp u(a_1) \rangle_c)^K \right) \right| \\ & \leq \mathbb{E} \left( \langle d + d^2(1 + \sqrt{|u(a_1)|})^2 \rangle (1 - \langle \exp u(a_1) \rangle_c)^K \right) \\ & \leq \sqrt{\mathbb{E} \langle [d + d^2(1 + \sqrt{|u(a_1)|})^2]^2 \rangle \mathbb{E} \langle (1 - \exp u(a_1))^{2K} \rangle_c} \\ & \leq \sqrt{C \mathbb{E} \langle (1 - \exp u(a_1))^{2K} \rangle_c},\end{aligned}$$

where in the first line we used Assumption 1, in the second Cauchy–Schwarz and Jensen’s inequalities, and in the last one that by Lemma C1 there exists a constant  $C > 0$  that bounds the first factor in the square root. Now, because  $(1 - \exp u(a_1))^{2K}$  is a continuous and bounded function of  $a_1$ , by Lemma C3 we have that

$$\mathbb{E} \langle (1 - \exp u(a_1))^{2K} \rangle_c \xrightarrow{m \rightarrow +\infty} \mathbb{E}(1 - \exp u(\theta))^{2K}.$$

By taking the limit  $m \rightarrow +\infty$  in equation (C.4) and using the convergence of the overlap  $Q_{1,1}$  along the subsequence  $(N_m)_{m \geq 1}$ , this implies that

$$\bar{r} = \alpha \sum_{k \leq K} \mathbb{E}f_k(\theta_1, \dots, \theta_k) + \mathcal{O} \left( \sqrt{\mathbb{E}(1 - \exp u(\theta))^{2K}} \right).$$

Consider the two terms in  $f_k$  separately. Taking the limit  $K \rightarrow +\infty$ , by the monotone convergence theorem,

$$\begin{aligned}\sum_{k \leq K} \mathbb{E}u'(\theta)^2 \exp u(\theta) \prod_{j=2}^k (1 - \exp u(\theta_j)) &= \mathbb{E}_{\tilde{z}} \left[ \mathbb{E}_{\xi} u'(\theta)^2 \exp u(\theta) \sum_{k \leq K} (1 - \mathbb{E}_{\xi} \exp u(\theta))^{k-1} \right] \\ &\xrightarrow{K \rightarrow +\infty} \mathbb{E}_{\tilde{z}} \frac{\mathbb{E}_{\xi} u'(\theta)^2 \exp u(\theta)}{\mathbb{E}_{\xi} \exp u(\theta)},\end{aligned}$$



and since  $|u''(a_1)| \leq d$  by Assumption 1, by the dominated convergence theorem, we also have

$$\sum_{k \leq K} \mathbb{E} u''(\theta) \exp u(\theta) \prod_{j=2}^k (1 - \exp u(\theta_j)) \xrightarrow{K \rightarrow +\infty} \mathbb{E}_{\tilde{z}} \frac{\mathbb{E}_{\xi} u'(\theta) \exp u(\theta)}{\mathbb{E}_{\xi} \exp u(\theta)}.$$

To see that the second term goes to 0, note that  $1 - \exp u(\theta) < 1$ . The limit then follows by the dominated convergence theorem. This proves  $\bar{r} = \bar{\Phi}(q, \rho)$ , i.e. equation (3.5).

The proof that  $r = \Phi(q, \rho)$  (equation (3.4)) follows in a similar way. Let  $K \geq 1$ . For each  $k \leq K$  define

$$g_k(a_1, \dots, a_{2k}) := u'(a_1)u'(a_2) \exp(u(a_1)u(a_2)) \prod_{j=2}^k (1 - \exp(u(a_{2j-1}) + u(a_{2j}))).$$

By Assumption 1, all these are continuous and bounded functions of  $a_1, \dots, a_{2k}$ . As before, by the cavity formula and the geometric series we have that

$$\mathbb{E} \langle Q_{1,2} \rangle = \alpha \sum_{k \leq K} \mathbb{E} \langle g_k(a_1, \dots, a_{2k}) \rangle_c + \alpha \mathbb{E} \left( \langle u'(a_1)u'(a_2) \rangle (1 - \langle \exp(u(a_1) + u(a_2)) \rangle_c)^K \right).$$

Finally, the conclusion follows in a similar manner as for the first equation: Lemma C3 is used for the sum while the second term can be seen to go to 0 when  $K \rightarrow +\infty$ .  $\square$

To establish the second pair of fixed point equations, a feasible approach would be to perform a cavity argument over the variables indexed by  $i = 1, \dots, N$  analogous to the proof of Proposition C1 (this ‘cavity in  $N$ ’ is in general more involved than the ‘cavity in  $M$ ’). Nevertheless, by the virtue of concentration of the overlaps, we show that they can also be obtained through an elementary derivation via Gaussian integration by parts.

**PROPOSITION C2.** The vector  $(\rho, q, \bar{r}, r)$  satisfies (3.2) and (3.3).

*Proof.* To see this, note that by the term  $-\kappa(\sigma_1^1)^2$  in the Hamiltonian (2.2) we can use Gaussian integration by parts with respect to  $\sigma_1^1$  and the Gaussian variables  $(g_{k1})_{k \leq M}$ . In particular

$$\mathbb{E} \langle f((\sigma^\ell), g_{ki}) g_{ki} \rangle = \mathbb{E} \langle \partial_{g_{ki}} f \rangle + \sum_{\ell} \left( \mathbb{E} \langle f \partial_{g_{ki}} H_N(\sigma^\ell) \rangle - \mathbb{E} \langle f \rangle \langle \partial_{g_{ki}} H_N(\sigma^\ell) \rangle \right),$$

and, denoting the modified Hamiltonian  $\hat{H}_N(\sigma) := \sum_{k \leq M} u(S_k(\sigma) + z_k) - h \sum_{i \leq N} \sigma_i$ ,

$$2\kappa \mathbb{E} \langle f(\sigma_i) \sigma_i \rangle = \mathbb{E} \langle \partial_{\sigma_i} f \rangle + \mathbb{E} \langle f \partial_{\sigma_i} \hat{H}_N(\sigma) \rangle.$$

These identities imply (recall the definitions (2.5))

$$\begin{aligned}
2\kappa \mathbb{E}\langle \sigma_1 \rangle &= \mathbb{E} \left\langle \frac{1}{\sqrt{N}} \sum_{k \leq M} A_k g_{k1} - h \right\rangle \\
&= \mathbb{E} \left\langle \frac{\sigma_1^1}{N} \sum_{k \leq M} \left( (A_k^1)^2 + B_k^1 - A_k^1 A_k^2 \right) - h \right\rangle, \\
2\kappa \mathbb{E}\langle (\sigma_1)^2 \rangle &= \mathbb{E} \left\langle 1 + \sigma_1 \left( \frac{1}{\sqrt{N}} \sum_{k \leq M} A_k g_{k1} - h \right) \right\rangle \\
&= \mathbb{E} \left\langle 1 + \frac{(\sigma_1^1)^2}{N} \sum_{k \leq M} \left( (A_k^1)^2 + B_k^1 \right) - \frac{\sigma_1^1 \sigma_1^2}{N} \sum_{k \leq M} A_k^1 A_k^2 - h \sigma_1 \right\rangle, \\
2\kappa \mathbb{E}\langle \sigma_1^1 \sigma_1^2 \rangle &= \mathbb{E} \left\langle \sigma_1^2 \left( \frac{1}{\sqrt{N}} \sum_{k \leq M} A_k^1 g_{k1} - h \right) \right\rangle \\
&= \mathbb{E} \left\langle \frac{\sigma_1^1 \sigma_1^2}{N} \sum_{k \leq M} \left( (A_k^1)^2 + B_k^1 \right) + \frac{(\sigma_1^2)^2}{N} \sum_{k \leq M} A_k^1 A_k^2 - 2 \frac{\sigma_1^2 \sigma_1^3}{N} \sum_{k \leq M} A_k^1 A_k^3 - h \sigma_1^2 \right\rangle.
\end{aligned}$$

The concentrations of  $R_{1,1}$ ,  $R_{1,2}$ ,  $Q_{1,1}$  and  $Q_{1,2}$  (see Proposition B1) then implies that

$$\begin{aligned}
\mathbb{E}\langle \sigma_1 \rangle &= -\frac{h}{2\kappa + r_m - \bar{r}_m} + o_m(1), \\
2\kappa \rho_m &= 1 + \rho_m \bar{r}_m - q_m r_m + \frac{h^2}{2\kappa + r_m - \bar{r}_m} + o_m(1), \\
2\kappa q_m &= -q_m(r_m - \bar{r}_m) + (\rho_m - q_m)r_m + \frac{h^2}{2\kappa + r_m - \bar{r}_m} + o_m(1).
\end{aligned}$$

Sending  $m \rightarrow +\infty$ , these lead to  $q = \Psi(r, \bar{r})$  and  $\rho = \bar{\Psi}(r, \bar{r})$  (equations (3.2) and (3.3)) and complete our proof.  $\square$

Propositions C1 and C2 together conclude that  $(\rho, q, \bar{r}, r)$  satisfies (3.2)–(3.5). We are now in position to prove Theorem 2.

### C.3 Proof of Theorem 2.

Consider any subsequence of system sizes. As explained above Lemma C3, there exists some further subsequence  $(N_m)_{n \geq 1}$  such that

$$(\rho_{N_m}, q_{N_m}, \bar{r}_{N_m}, r_{N_m}) \xrightarrow{m \rightarrow +\infty} (\rho, q, \bar{r}, r)$$

for some  $(\rho, q, \bar{r}, r)$  in a compact set  $K$ . By Propositions C1 and C2 and the continuity of the functions  $\bar{\Phi}$ ,  $\Phi$ ,  $\bar{\Psi}$  and  $\Psi$ , we see that  $(\rho, q, \bar{r}, r)$  is a solution of (3.2)–(3.5). Since this system of equations has

only one unique solution as ensured by the given assumption, it follows that any convergent subsequence will share the same limit and this implies the assertion.

#### D. Proof of Theorem 1

In this section, we establish the convergence of  $F_N$  using a basic interpolation in this section. Let  $\alpha, \Delta_*, \Delta, \kappa > 0$  be fixed. For  $t \in [0, 1]$ , consider an auxiliary free energy

$$\phi_N(t) := \frac{1}{N} \mathbb{E} \ln \int \exp \left( \sum_{k \leq M} u(\sqrt{t}((\bar{G}\sigma)_k + z_k)) - h \sum_{i \leq N} \sigma_i - \kappa \|\sigma\|^2 \right) d\sigma.$$

Let  $\mathbb{E}\langle \cdot \rangle_t$  be the joint mean with respect to the Gaussian  $(G, z)$  and Gibbs measure associated with the Hamiltonian (2.2) after replacing  $u(x)$  by  $u(\sqrt{t}x)$ . We need the following crucial lemma.

**LEMMA D1.** For any  $t_0 \in (0, 1]$ , there exist constants  $K > 0$  and  $\delta > 0$  such that for all  $t \in [t_0 - \delta, t_0 + \delta] \cap (0, 1]$  and  $N \geq 1$  we have

$$\max \left( \left| \frac{d}{dt} \mathbb{E}\langle R_{1,1} \rangle_t \right|, \left| \frac{d}{dt} \mathbb{E}\langle R_{1,2} \rangle_t \right|, \left| \frac{d}{dt} \mathbb{E}\langle Q_{1,1} \rangle_t \right|, \left| \frac{d}{dt} \mathbb{E}\langle Q_{1,2} \rangle_t \right| \right) \leq K.$$

*Proof.* We establish this inequality only for  $\frac{d}{dt} \mathbb{E}\langle R_{1,1} \rangle_t$  as the others can be treated similarly. Let

$$L(\sigma) := \frac{1}{N} \sum_{k \leq M} (S_k(\sigma) + z_k) u'(\sqrt{t}(S_k(\sigma) + z_k)).$$

Write

$$\begin{aligned} \frac{d}{dt} \mathbb{E}\langle R_{1,1} \rangle_t &= \frac{N}{2\sqrt{t}} \mathbb{E}\langle L(\sigma^1) - L(\sigma^2) \rangle_t \\ &= \frac{N}{2\sqrt{t}} \mathbb{E}\langle (R_{1,1} - \langle R_{1,1} \rangle_t)(L(\sigma^1) - \langle L(\sigma^1) \rangle_t) \rangle_t. \end{aligned}$$

As in Appendix B.4,

$$\mathbb{E}\langle (R_{1,1} - \langle R_{1,1} \rangle_t)^2 \rangle_t \leq \frac{K}{N} \quad \text{and} \quad \mathbb{E}\langle (L(\sigma^1) - \langle L(\sigma^1) \rangle_t)^2 \rangle_t \leq \frac{K}{N},$$

so, by the Cauchy–Schwarz inequality,  $\frac{d}{dt} \mathbb{E}\langle R_{1,1} \rangle_t \leq K$ . Since this bound is valid locally uniformly in  $t$ , the assertion follows.  $\square$

We now turn to the proof of Theorem 1. Let  $(q, \rho, \bar{r}, r)$  be the unique solution to (3.2)–(3.5). As explained in Appendix 3,  $F(q, \rho, r, \bar{r}) = F(q, \rho)$ . Also, from Lemma B8,  $\lim_{N \rightarrow \infty} \text{Var}(F_N) = 0$ . Our proof will be complete if we can show that all subsequential limits of  $\mathbb{E}F_N$  are equal to  $F(q, \rho, r, \bar{r})$ . We establish this statement as follows.

First of all, from Lemma D1, we can use the Arzela–Ascoli theorem and a diagonalization process to further pass to a subsequence along which  $\mathbb{E}\langle R_{1,1} \rangle_t$ ,  $\mathbb{E}\langle R_{1,2} \rangle_t$ ,  $\mathbb{E}\langle Q_{1,1} \rangle_t$  and  $\mathbb{E}\langle Q_{1,2} \rangle_t$  converge to some  $\rho_t, q_t, \bar{r}_t, r_t$  uniformly with respect to  $t \in [0, 1]$ , respectively. For the rest of the proof, to lighten our notation, we shall simply assume that the convergences of these overlaps and  $\mathbb{E}F_N$  are valid along the original sequence  $N$ . Here,  $\rho_t, q_t, \bar{r}_t, r_t$  are defined on  $t \in [0, 1]$  and are locally Lipschitz on  $(0, 1]$ . Furthermore, from Theorem 2, they satisfy (3.2)–(3.5) for  $u(x)$  being replaced by  $u(\sqrt{t}x)$  for all  $t \in [0, 1]$ .

Now, from the uniqueness of the solution to the fixed point equations (3.2)–(3.5) (corresponding to  $t = 1$ ),

$$(\rho_1, q_1, \bar{r}_1, r_1) = (\rho, q, \bar{r}, r). \quad (\text{D.1})$$

When  $t = 0$ ,

$$\rho_0 = \frac{1}{2\kappa} + \frac{h^2}{(2\kappa)^2}, \quad q_0 = \frac{h^2}{2\kappa}, \quad \bar{r}_0 = 0 \text{ and } r_0 = 0. \quad (\text{D.2})$$

Sending  $t \downarrow 0$  in (3.2)–(3.5) also yields that  $\lim_{t \downarrow 0} (\rho_t, q_t, \bar{r}_t, r_t) = (\rho_0, q_0, \bar{r}_0, r_0)$ . Hence,  $\rho_t, q_t, \bar{r}_t, r_t$  are continuous on  $[0, 1]$ .

Next, set

$$f(t) := \alpha \mathbb{E} \ln \mathbb{E}_\xi \exp u(\sqrt{t}\theta_t) + \frac{1}{2} \left( \frac{r_t + h^2}{2\kappa + r_t - \bar{r}_t} - \ln(2\kappa + r_t - \bar{r}_t) + r_t q_t - \bar{r}_t \rho_t + \ln 2\pi \right),$$

where  $\theta_t := \tilde{z}\sqrt{\Delta_* + q_t} + \xi\sqrt{\rho_t - q_t}$  with i.i.d. standard Gaussian  $\tilde{z}, \xi$ . Because the fixed point equations satisfy the critical point conditions

$$\frac{\partial f}{\partial \rho_t} = \frac{\partial f}{\partial q_t} = \frac{\partial f}{\partial \bar{r}_t} = \frac{\partial f}{\partial r_t} = 0$$

and  $\rho_t, q_t, \bar{r}_t, r_t$  are locally Lipschitz on  $(0, 1)$ , we conclude that  $df/dt = \partial f/\partial t$ . To compute this partial derivative, let  $\theta'_t := \tilde{z}\sqrt{\Delta_* + q_t} + \xi'\sqrt{\rho_t - q_t}$  for  $\xi'$  an independent copy of  $\xi$ . Using Gaussian integration by parts with respect to  $\tilde{z}$  and  $\xi$  yields

$$f'(t) = \frac{\partial f}{\partial t} = \frac{\alpha}{2\sqrt{t}} \mathbb{E}_{\tilde{z}} \frac{\mathbb{E}_\xi u'(\sqrt{t}\theta_t) \exp u(\sqrt{t}\theta_t) \theta_t}{\mathbb{E}_{\xi'} u'(\sqrt{t}\theta'_t) \exp u(\sqrt{t}\theta'_t)} = \frac{1}{2} (\bar{r}_t(\rho_t + \Delta_*) - r_t(q_t + \Delta_*)).$$

On the other hand, note that

$$\phi'_N(t) = \frac{1}{2} \mathbb{E} \langle Q_{1,1}(R_{1,1} + \Delta_*) - Q_{1,2}(R_{1,2} + \Delta_*) \rangle_t, \quad (\text{D.3})$$

where we used Gaussian integration by parts with respect to the elements of  $\tilde{G}$  and  $z$  similar to the proof of Proposition C2. By Theorem 2, for every  $t \in [0, 1]$ ,  $(R_{1,1}, R_{1,2}, Q_{1,1}, Q_{1,2})$  converges to  $(\rho_t, q_t, \bar{r}_t, r_t)$  under  $\mathbb{E}\langle \cdot \rangle_t$  and, hence,

$$\lim_{N \rightarrow +\infty} \phi'_N(t) = \frac{1}{2} (\bar{r}_t(\rho_t + \Delta_*) - r_t(q_t + \Delta_*)).$$

To complete the proof, note that  $\phi_N(1) = \mathbb{E}F_N$  and that from (D.1) and (D.2),

$$f(0) = \alpha u(0) + \frac{h^2}{4\kappa} + \frac{1}{2} \ln \frac{\pi}{\kappa} = \phi_N(0),$$

$$f(1) = F(q, \rho, \bar{r}, r).$$

Consequently, from (D.3) and noting that  $\mathbb{E}\langle R_{1,1} \rangle_t, \mathbb{E}\langle R_{1,2} \rangle_t, \mathbb{E}\langle Q_{1,1} \rangle_t$  and  $\mathbb{E}\langle Q_{1,2} \rangle_t$  are uniformly bounded on  $[0, 1]$  due to Lemma B6, the dominated convergence theorem implies

$$|\mathbb{E}F_N - F(q, \rho, \bar{r}, r)| = |\phi_N(1) - f(1)| \leq \int_0^1 |\phi'_N(t) - f'(t)| dt \rightarrow 0,$$

completing our proof.  $\square$

### E. Proof of Theorem 3

Throughout the entire section, the assumptions in Theorem 3 are in force. Let  $h_N = 2\kappa\sqrt{\gamma_N}$  and  $h = 2\kappa\sqrt{\gamma}$ . From the convergence (3.7),

$$\lim_{N \rightarrow \infty} \mathbb{E}|h_N - h|^2 = 0.$$

Recall  $F_N$  from (2.3). Throughout this section, we denote by  $F_N(h), F_N(h_N)$  and  $\langle \cdot \rangle_h, \langle \cdot \rangle_{h_N}$  the free energies  $F_N$  and Gibbs expectations  $\langle \cdot \rangle$  associated with the deterministic  $h$  and randomized  $h_N$ , respectively. In the latter case, we also assume that  $h_N$  is independent of all other randomness. First of all, we establish the following lemma:

LEMMA E1.

$$\lim_{N \rightarrow \infty} \mathbb{E}|F_N(h_N) - F_N(h)|^2 = 0 \quad (\text{E.1})$$

and

$$\lim_{N \rightarrow \infty} \mathbb{E}\langle R_{1,2} \rangle_{h_N} = \lim_{N \rightarrow \infty} \mathbb{E}\langle R_{1,2} \rangle_h = q. \quad (\text{E.2})$$

*Proof.* For  $0 \leq t \leq 1$ , consider the interpolated free energy

$$F_{N,t} = \frac{1}{N} \log \int \exp \left( \sum_{k=1}^M u(S_k(\sigma) + z_k) - (th_N + (1-t)h) \sum_{i=1}^N \sigma_i - \kappa \|\sigma\|^2 \right) d\sigma,$$

and denote by  $\langle \cdot \rangle_t$  the Gibbs expectation associated with this free energy. Clearly  $F_{N,1} = F_N(h_N)$  and  $F_{N,0} = F_N(h)$ . A direct differentiation gives that

$$\frac{d}{dt} F_{N,t} = (h - h_N) \langle M_1 \rangle_t$$

and

$$\begin{aligned} \frac{d}{dt} \langle R_{1,2} \rangle_t &= 2N(h - h_N) \langle R_{1,2}(M_1 - M_3) \rangle_t \\ &= 2N(h - h_N) \langle (R_{1,2} - \langle R_{1,2} \rangle_t)(M_1 - \langle M_1 \rangle_t) \rangle_t, \end{aligned}$$

where  $M_\ell := N^{-1} \sum_{i=1}^N \sigma_i^\ell$ . From these and the Hölder inequality,

$$\mathbb{E}|F_N(h_N) - F_N(h)|^2 \leq \left( \mathbb{E}|h_N - h|^2 \right)^{1/2} \left( \mathbb{E}\langle M_1^2 \rangle_t \right)^{1/2} \quad (\text{E.3})$$

$$\leq \left( \mathbb{E}|h_N - h|^2 \right)^{1/2} \left( \frac{\mathbb{E}\langle \|\sigma^1\|^2 \rangle_t}{N} \right)^{1/2} \quad (\text{E.4})$$

and

$$\begin{aligned} &|\mathbb{E}\langle R_{1,2} \rangle_{h_N} - \mathbb{E}\langle R_{1,2} \rangle_h| \\ &\leq 2N \left( \mathbb{E}|h_N - h|^2 \right)^{1/2} \left( \mathbb{E}\langle |R_{1,2} - \langle R_{1,2} \rangle_t|^4 \rangle_t \right)^{1/4} \left( \mathbb{E}\langle |M_1 - \langle M_1 \rangle_t|^4 \rangle_t \right)^{1/4}. \end{aligned} \quad (\text{E.5})$$

Recall from Lemma B3, the same proof yields that for any  $k \geq 1$ , there exists some  $K > 0$  such that for all  $t \in [0, 1]$  and  $N \geq 1$ ,

$$\langle \|\sigma\|^{2k} \rangle_t \leq KN^k \left( \|\bar{z}\|^{2k} + \|\bar{G}\|_{op}^{2k} + h^{2k} + h_N^{2k} + 1 \right). \quad (\text{E.6})$$

Since the expectation of every term in the bracket is bounded by a universal constant, we obtain (E.1) by using (E.3) and taking  $k = 1$ .

As for (E.2), note that for any fixed  $w \in \mathbb{R}^N$ ,  $f(\sigma) := N^{-1}w \cdot \sigma$  is Lipschitz with Lipschitz constant  $\|w\|/N$  and that the Hessian of the interpolated Hamiltonian in  $F_{N,t}$  is bounded from above by  $-\kappa I/2$ . From the Brascamp–Lieb inequality [12, theorem 5.1],

$$\left\langle \left| \frac{w \cdot \sigma^1}{N} - \frac{w \cdot \langle \sigma^1 \rangle_t}{N} \right|^4 \right\rangle_t \leq \frac{K\|w\|^4}{N^4}, \quad (\text{E.7})$$

where  $K$  is a constant independent of  $w, N, t$ . Therefore, letting  $w = (1, \dots, 1)$  implies

$$\langle |M_1 - \langle M_1 \rangle_t|^4 \rangle_t \leq \frac{K\|w\|^4}{N^4} = \frac{K}{N^2}. \quad (\text{E.8})$$

In addition, writing

$$R_{1,2} - \langle R_{1,2} \rangle_t = \frac{\sigma^1 \cdot \sigma^2}{N} - \frac{\langle \sigma^1 \rangle_t \cdot \sigma^2}{N} + \frac{\langle \sigma^1 \rangle_t \cdot \sigma^2}{N} - \frac{\langle \sigma^1 \rangle_t \cdot \langle \sigma^2 \rangle_t}{N},$$

we can apply (E.7) with the choices  $w = \sigma^2$  and  $w = \langle \sigma^1 \rangle_t$  to get

$$\left\langle \left| \frac{\sigma^1 \cdot \sigma^2}{N} - \frac{\langle \sigma^1 \rangle_t \cdot \sigma^2}{N} \right|^4 \right\rangle_t \leq \frac{K\|\sigma^2\|^4}{N^4} = \frac{K\langle \|\sigma^1\|^4 \rangle_t}{N^4}$$

and

$$\left\langle \left| \frac{\langle \sigma^1 \rangle_t \cdot \sigma^2}{N} - \frac{\langle \sigma^1 \rangle_t \cdot \langle \sigma^2 \rangle_t}{N} \right|^4 \right\rangle_t \leq \frac{K\|\langle \sigma^1 \rangle_t\|^4}{N^4} \leq \frac{K\langle \|\sigma^1\|^4 \rangle_t}{N^4}.$$

Putting these together and applying (E.6) with  $k = 2$ , we obtain that for all  $0 \leq t \leq 1$  and  $N \geq 1$ ,

$$\mathbb{E} \langle |R_{1,2} - \langle R_{1,2} \rangle_t|^4 \rangle_t \leq \frac{32K\mathbb{E} \langle \|\sigma^1\|^4 \rangle_t}{N^4} \leq \frac{K'}{N^2} \quad (\text{E.9})$$

for some constant  $K'$  independent of  $t, N$ . Now plugging (E.8) and (E.9) into (E.5) validates (E.2).  $\square$

Now we proceed to establish (3.8) and (3.9). From Theorem 2 and (A.3),

$$\lim_{N \rightarrow \infty} \mathbb{E} \langle R_{1,2} \rangle_h = q$$

and

$$\frac{1}{N} \mathbb{E} \langle \|x\|^2 - x^* \rangle^2 = \frac{1}{N} \mathbb{E} \langle \|\sigma\|_{h_N}^2 \rangle^2 = \mathbb{E} \langle R_{1,2} \rangle_{h_N}.$$

These and (E.2) together yields (3.9). As for (3.8), from (A.1), we can write

$$\tilde{F}_N = -\kappa \gamma_N + F_N(x^*),$$

where

$$F_N(x^*) := \frac{1}{N} \ln \int_{\mathbb{R}^N} \exp \left( \sum_{k \leq M} u \left( (\bar{G} O^T \sigma)_k - z_k \right) - h_N \sum_{i \leq N} \sigma_i - \kappa \|\sigma\|^2 \right) d\sigma$$

and  $O$  is an orthonormal matrix satisfying  $Ox^* = N^{-1/2} \|x^*\| \mathbf{1}_N$ . Write

$$\begin{aligned} \tilde{F}_N - \mathbb{E}\tilde{F}_N &= -\kappa(\gamma_N - \mathbb{E}\gamma_N) \\ &\quad + F_N(x^*) - \mathbb{E}[F_N(x^*)|x^*] \\ &\quad + \mathbb{E}[F_N(x^*)|x^*] - \mathbb{E}[F_N(x^*)]. \end{aligned} \tag{E.10}$$

Here the first term on the right-hand side vanishes in  $L^2$  norm by the given assumption. For the second term, observe that conditionally on  $x^*$ ,  $F_N(x^*)$  is the same as  $F_N(h_N)$  in distribution. In addition, since  $h_N$  is independent of  $(G, z)$  and the term  $h_N \sum_{i=1}^N \sigma_i$  in the Hamiltonian of  $F_N(h_N)$  is linear, it can be checked that the proof of Lemma B8 is independent of whether  $h$  is deterministic or randomized so that we can still apply the conclusion of Lemma B8 (with  $n = 1$  and  $\lambda = \eta = 0$ ) to get that with probability one,

$$\text{Var}(F_N(x^*)|x^*) = \text{Var}(F_N(h_N)|h_N) \leq \frac{K}{N},$$

where  $K$  is independent of  $N$ . Hence, we can bound the second term by

$$\mathbb{E}|F_N(x^*) - \mathbb{E}[F_N(x^*)|x^*]|^2 = \mathbb{E}[\text{Var}(F_N(x^*)|x^*)] \leq \frac{K}{N}.$$

Lastly, write

$$\begin{aligned} \mathbb{E}[F_N(x^*)|x^*] - \mathbb{E}[F_N(x^*)] &= \mathbb{E}[F_N(h_N)|h_N] - \mathbb{E}[F_N(h_N)] \\ &= \mathbb{E}[F_N(h_N) - F_N(h)|h_N] \\ &\quad + (\mathbb{E}[F_N(h)] - \mathbb{E}[F_N(h_N)]). \end{aligned}$$

We bound the third term in (E.10) by using the Jensen inequality,

$$\begin{aligned} \mathbb{E}|\mathbb{E}[F_N(x^*)|x^*] - \mathbb{E}[F_N(x^*)]|^2 &\leq 4\mathbb{E}|\mathbb{E}[F_N(h_N) - F_N(h)|h_N]|^2 \\ &\quad + 4(\mathbb{E}[F_N(h)] - \mathbb{E}[F_N(h_N)])^2 \\ &\leq 8\mathbb{E}|F_N(h_N) - F_N(h)|^2 \rightarrow 0, \end{aligned}$$

where the last limit used (E.1). This completes our proof for (3.8).